

EXAMPLE 1.7.1. The airfoil section with 4 percent maximum camber, which is described in Example 1.6.1, has a sealed 20 percent trailing-edge flap. Estimate the section flap effectiveness and the section lift coefficient for a 5-degree flap deflection and a 0-degree angle of attack relative to the undeflected chord line.

Solution. From Eq. (1.7.5), a flap fraction of 0.20 gives

$$\theta_f = \cos^{-1}\left(2\frac{c_f}{c}-1\right) = \cos^{-1}[2(0.20)-1] = 2.214297$$

Using this in Eq. (1.7.6) yields an ideal section flap effectiveness of

$$\varepsilon_{fi} = 1 - \frac{\theta_f - \sin \theta_f}{\pi} = 1 - \frac{2.214297 - \sin(2.214297)}{\pi} = 0.55$$

Applying this and results from Figs. 1.7.4 and 1.7.5 to Eq. (1.7.13) yields

$$\varepsilon_f = \eta_h \eta_d \varepsilon_{fi} = (0.86)(1.00)(0.55) = \underline{0.47}$$

With this and the value of α_{L0} from Example 1.6.1, Eq. (1.7.12) results in

$$\tilde{C}_L = \tilde{C}_{L,\alpha}[\alpha - \alpha_{L0}(0) + \varepsilon_f \delta] = 2\pi[0.0 - (-0.08) + 0.47(5\pi/180)] = \underline{0.76}$$

1.8. Incompressible Flow over Finite Wings

In Sec. 1.6 we reviewed the aerodynamic properties of airfoils, which are the same as the aerodynamic properties of wings with infinite span. A wing of constant cross-section and infinite span would have no variation in aerodynamic forces in the spanwise direction. The aerodynamic forces per unit span acting on such a wing, at any point along the span, would be the same as those for an airfoil of the same cross-section. An airfoil or wing of infinite span is synthesized using vortex sheets that are made up of straight vortex filaments that extend to $\pm\infty$ in the direction of span. The vortex strength can vary over the sheet as we move in a chordwise direction from one vortex filament to another. However, there is no variation in vortex strength as we move along a vortex filament in the direction of span.

Any real wing, of course, must have finite span. At the tips of a finite wing, the air on the lower surface of the wing comes in direct contact with the air on the upper surface of the wing. Thus, at the tips of a finite wing, the pressure difference between the upper and lower surfaces must always go to zero. As a result, the lift on any finite wing must go to zero at the wingtips, as shown schematically in Fig. 1.8.1.

The pressure difference between the upper and lower surfaces of a finite wing is reduced near the wingtips, because some of the air from the high-pressure region below the wing spills outward, around the wingtip, and back inward toward the low-pressure region above the wing. Thus, while the flow around an infinite wing is entirely in the plane of the airfoil section, the flow around a finite wing is three-dimensional.

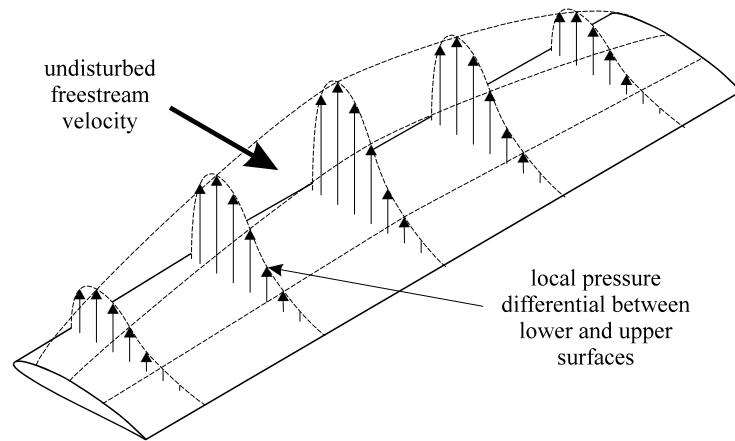


Figure 1.8.1. Lift distribution on a finite wing.

As air flows over a finite wing, the air below the wing moves outward toward the wingtip and the air above the wing moves inward toward the root, as shown in Fig. 1.8.2. Where the flows from the upper and lower surfaces combine at the trailing edge, the difference in spanwise velocity generates a trailing vortex sheet. Because this vortex sheet is free and not bound to the wing's surface, the flow field induced by the sheet tends to change the shape of the sheet as it moves downstream from the wing. As can be seen in Fig. 1.8.3, this vortex sheet rolls up around an axis trailing slightly inboard from each wingtip. At some distance behind the wing, the sheet becomes completely rolled up to form two large vortices, one trailing aft of each wingtip. For this reason,

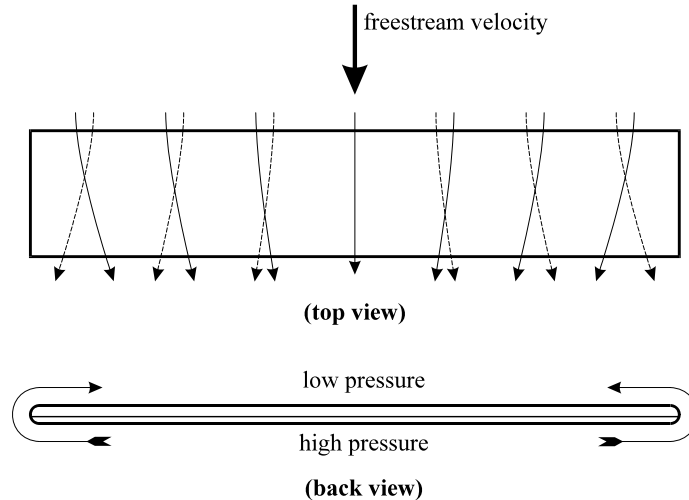


Figure 1.8.2. Airflow around a finite wing.

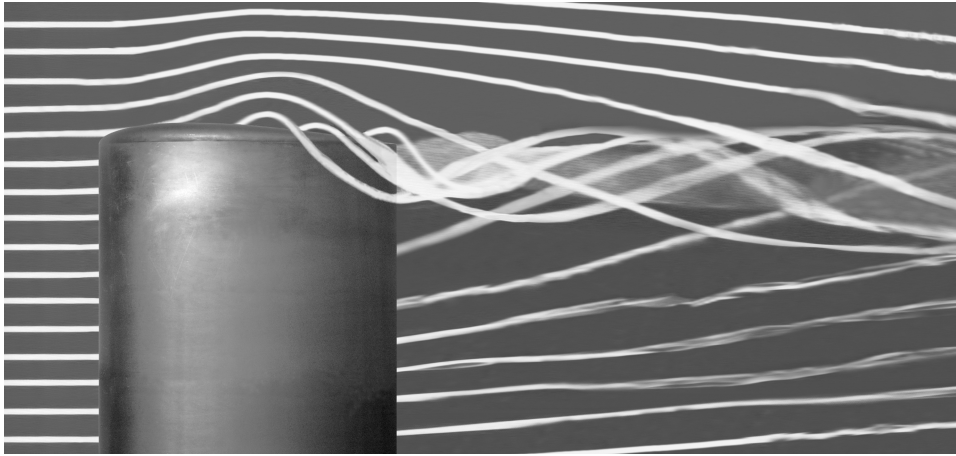


Figure 1.8.3. Vorticity trailing aft of a finite wing.

these vortices are referred to as *wingtip vortices*, even though they are generated over the full span of the wing. The downward velocity component that is induced between the wingtip vortices is called *downwash*. Potential flow theory predicts that the wingtip vortices must trail behind the wing for an infinite distance with no reduction in strength. In reality, viscous effects will eventually dissipate the energy in these vortices, but this is a slow process. These vortices will still have significant energy several miles behind a large aircraft and are of sufficient strength to cause control loss or structural damage to other aircraft following too closely.

From the discussion above, we see that a wing of finite span cannot be synthesized with a vortex sheet made up of vortex filaments that are always perpendicular to the airfoil sections of the wing. We can, however, still synthesize a finite wing with a vortex sheet, but the sheet must be made up of horseshoe-shaped vortex filaments. These filaments run out along the wing in the direction of span, curving back and eventually leaving the wing from the trailing edge at some point inboard of the wingtip. This is seen schematically in the plan view of the elliptic wing shown in Fig. 1.8.4. Because the strength of a vortex filament cannot vary along its length, the decrease in the circulation about the wing as we move out from the root toward the tip must result entirely from the vorticity that leaves the wing in the trailing vortex sheet.

The classical lifting-line theory developed by Prandtl (1918) was the first analytical method to satisfactorily predict the performance of a finite wing. While a general 3-D vortex lifting law was not available at the time of Prandtl's development, the 2-D vortex lifting law of Kutta (1902) and Joukowski (1906) was well known. Prandtl's lifting-line theory is based on the hypothesis that each spanwise section of a finite wing has a section lift that is equivalent to that acting on a similar section of an infinite wing having the same section circulation. However, to reconcile this theory with reality, the undisturbed freestream velocity in the Kutta-Joukowski law was intuitively replaced with the vector sum of the freestream velocity and the velocity induced by the trailing vortex sheet.

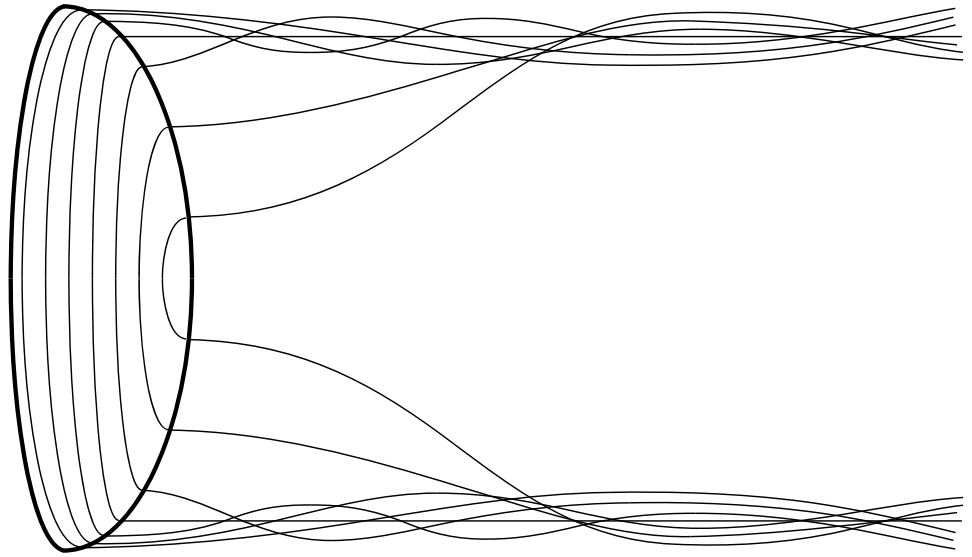


Figure 1.8.4. Schematic of the vorticity distribution on a finite wing with elliptic planform shape.

Today we know that Prandtl's hypothesis was correct. This can be shown as a direct consequence of the **vortex lifting law**, which is a 3-D counterpart to the 2-D Kutta-Joukowski law. This vortex lifting law states that for any potential flow containing vortex filaments, the force per unit length exerted on the surroundings at any point along a vortex filament is given by

$$d\mathbf{F} = \rho \Gamma \mathbf{V} \times d\mathbf{l} \quad (1.8.1)$$

where $d\mathbf{F}$ is the differential aerodynamic force vector and ρ , Γ , $\mathbf{V}$, and $d\mathbf{l}$ are, respectively, the fluid density, vortex strength, local fluid velocity, and the directed differential vortex length vector (see Saffman 1992).

The force computed from Eq. (1.8.1) is called the *vortex force* or *Kutta lift*. The vortex lifting law is a very useful tool in the study of aerodynamics and flight mechanics. It provides the basis for much of finite wing theory. There are two important consequences of Eq. (1.8.1). First, **for a bound vortex filament, the vortex force is always perpendicular to both the local velocity vector and the vortex filament**. Second, since a free vortex filament can support no force, **free vortex filaments must be aligned everywhere with the streamlines of the flow**.

Prandtl's Classical Lifting-Line Theory

Prandtl's lifting-line theory gives good agreement with experimental data for straight wings of aspect ratio greater than about 4. Development of this theory is presented in any undergraduate engineering textbook on aerodynamics and will not be repeated here. Only summary results are presented in this section.

The model used by Prandtl to approximate the bound vorticity and trailing vortex sheet is shown in Fig. 1.8.5. All bound vortex filaments are assumed to follow the wing quarter-chord line, and all trailing vortex filaments are assumed to be straight and parallel with the freestream. Rollup of the trailing vortex sheet is ignored.

The foundation of lifting-line theory is the requirement that for each cross-section of the wing, the lift predicted from the vortex lifting law must be equal to that predicted from airfoil section theory, i.e.,

$$\tilde{C}_L(z) = \tilde{C}_{L,\alpha}[\alpha_{\text{eff}}(z) - \alpha_{L0}(z)]$$

where α_{eff} is the local section angle of attack, including the effects of velocity induced by the trailing vortex sheet. Because of the downwash induced on the wing by the trailing vortex sheet, the local relative wind is inclined at an angle, α_i , to the freestream, as shown in Fig. 1.8.6. This angle is called the *induced angle of attack*. Since the lift is always perpendicular to the local relative wind, the downwash tilts the lift vector back, creating a component of lift parallel to the freestream. This is called *induced drag*.

When the downwash is accounted for, Prandtl's hypothesis requires that

$$\frac{2\Gamma(z)}{V_\infty c(z)} + \frac{\tilde{C}_{L,\alpha}}{4\pi V_\infty} \int_{\zeta=-b/2}^{b/2} \frac{1}{z-\zeta} \left(\frac{d\Gamma}{d\zeta} \right)_{\zeta=\zeta} d\zeta = \tilde{C}_{L,\alpha}[\alpha(z) - \alpha_{L0}(z)] \quad (1.8.2)$$

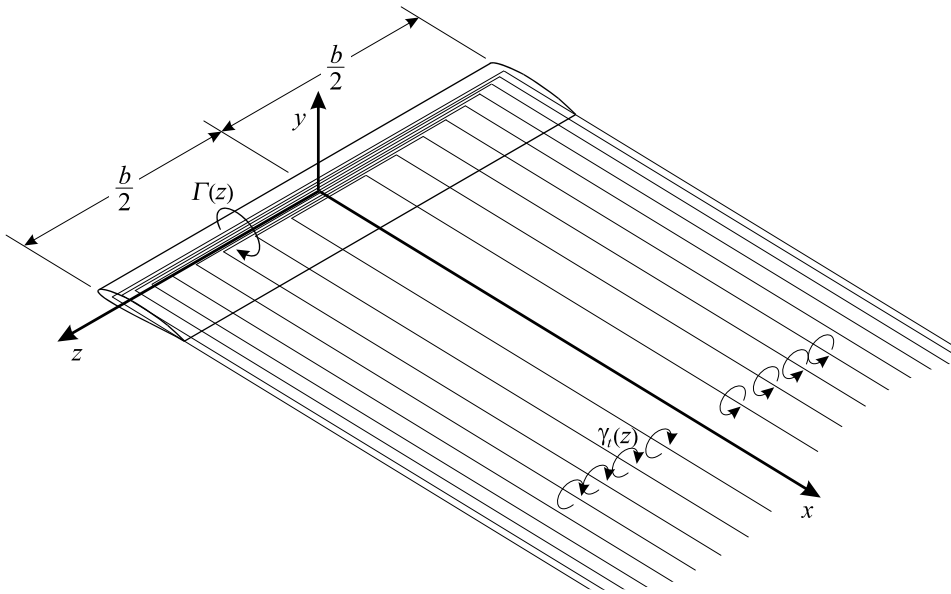


Figure 1.8.5. Prandtl's model for the bound vorticity and the trailing vortex sheet generated by a wing of finite span.

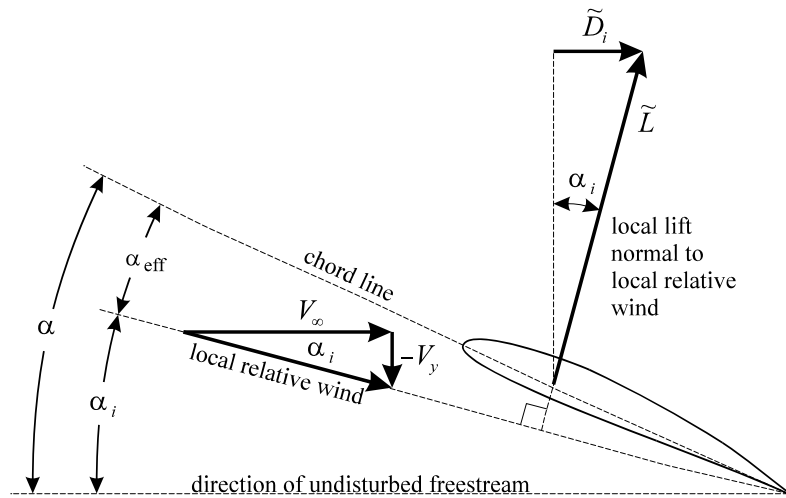


Figure 1.8.6. Induced angle of attack.

Equation (1.8.2) is the fundamental equation of Prandtl's lifting-line theory. It is a non-linear integrodifferential equation that involves only one unknown, the local section circulation as a function of the spanwise position, $\Gamma(z)$. All other parameters are known for a given wing design at a given geometric angle of attack and a given freestream velocity. Remember that the chord length, c , the geometric angle of attack, α , and the zero-lift angle of attack, α_{L0} , are all allowed to vary in the spanwise direction. The section lift slope could also vary with z but is usually assumed to be constant.

There is certain terminology associated with spanwise variation in wing geometry with which the reader should be familiar. If a wing has spanwise variation in geometric angle of attack as shown in Fig. 1.8.7, the wing is said to have *geometric twist*. The tip is commonly at a lower angle of attack than the root, in which case the geometric twist is referred to as *washout*. A spanwise variation in zero-lift angle of attack, like that shown in Fig. 1.8.8, is called *aerodynamic twist*. Deflecting a trailing-edge flap that extends over only part of the span is another form of aerodynamic twist (see Fig. 1.8.9).

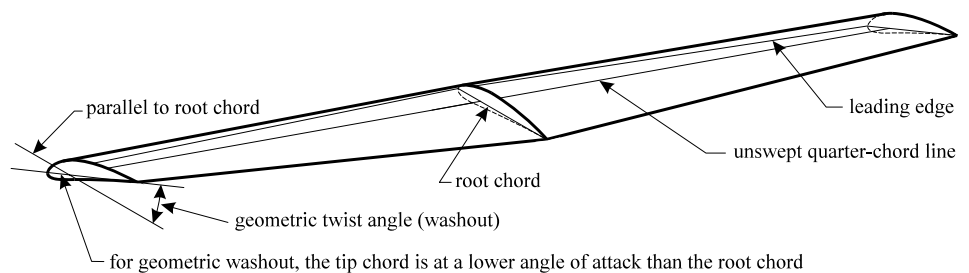


Figure 1.8.7. Geometric twist in an unswept rectangular wing.

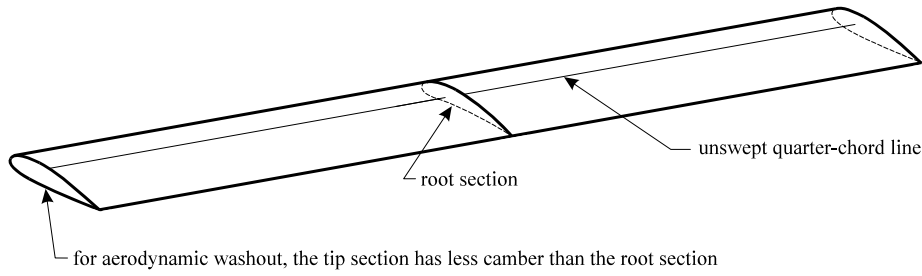


Figure 1.8.8. Aerodynamic twist in an unswept rectangular wing.

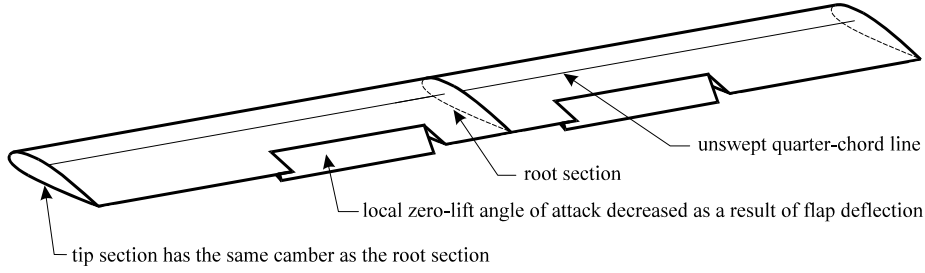


Figure 1.8.9. Aerodynamic twist resulting from deflection of a trailing-edge flap spanning only a portion of the wing.

For a single finite wing with no sweep or dihedral, an analytical solution to Prandtl's lifting-line equation can be obtained in terms of a Fourier sine series. For a given wing design, at a given angle of attack, the planform shape, $c(z)$, the geometric twist, $\alpha(z)$, and the aerodynamic twist, $\alpha_{L0}(z)$, are all known as functions of spanwise position. The circulation distribution is written as a Fourier series, the series is truncated to a finite number of terms (N), and the Fourier coefficients are determined by forcing the lifting-line equation to be satisfied at N specific sections along the span of the wing. From this solution the circulation distribution is given by

$$\Gamma(\theta) = 2bV_\infty \sum_{n=1}^N A_n \sin(n\theta); \quad \theta = \cos^{-1}(-2z/b) \quad (1.8.3)$$

where the Fourier coefficients, A_n , are obtained from

$$\sum_{n=1}^N A_n \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = \alpha(\theta) - \alpha_{L0}(\theta) \quad (1.8.4)$$

Once the circulation distribution has been determined, the section lift distribution can be obtained from the vortex lifting law. The resulting lift and induced drag coefficients for the finite wing are

$$C_L = \pi R_A A_1, \quad R_A \equiv \frac{b^2}{S} \quad (1.8.5)$$

$$C_{D_i} = \pi R_A \sum_{n=1}^N n A_n^2 = \frac{C_L^2}{\pi R_A e_s}, \quad e_s \equiv \frac{1}{1+\sigma}, \quad \sigma \equiv \sum_{n=2}^N n \left(\frac{A_n}{A_1} \right)^2 \quad (1.8.6)$$

The wingspan squared divided by the planform area, R_A , is called the *aspect ratio* and the parameter e_s is called the *span efficiency factor*.

Lifting-line theory predicts that an elliptic wing with no geometric or aerodynamic twist produces minimum possible induced drag for a given lift coefficient and aspect ratio. This planform has a chord that varies with the spanwise coordinate according to

$$c(z) = \frac{4b}{\pi R_A} \sqrt{1 - (2z/b)^2} \quad \text{or} \quad c(\theta) = \frac{4b}{\pi R_A} \sin(\theta)$$

As shown in Fig. 1.8.10, an unswept elliptic wing has a straight quarter-chord line. This gives the leading edge less curvature and the trailing edge more curvature than a conventional ellipse, which has a straight half-chord line. Several aircraft have been designed and built with elliptic wings. One of the best known is the British Spitfire, shown in Fig. 1.8.11.

The lift and induced drag coefficients predicted from Eqs. (1.8.5) and (1.8.6) for an **elliptic wing with no geometric or aerodynamic twist** are

$$C_L = C_{L,\alpha}(\alpha - \alpha_{L0}), \quad C_{L,\alpha} = \frac{\tilde{C}_{L,\alpha}}{1 + \tilde{C}_{L,\alpha}/(\pi R_A)} \quad (1.8.7)$$

$$C_{D_i} = \frac{C_L^2}{\pi R_A} \quad (1.8.8)$$

where $C_{L,\alpha}$ is the lift slope for the finite wing. Equation (1.8.7) shows that the lift slope for an untwisted elliptic wing is less than the section lift slope for the airfoil from which the wing was generated. However, as the aspect ratio for the wing becomes large, the

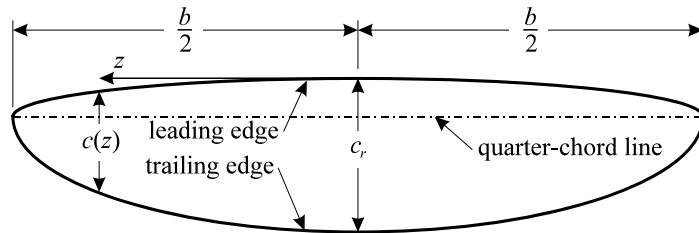


Figure 1.8.10. Planform shape of an unswept elliptic wing with an aspect ratio of 6.

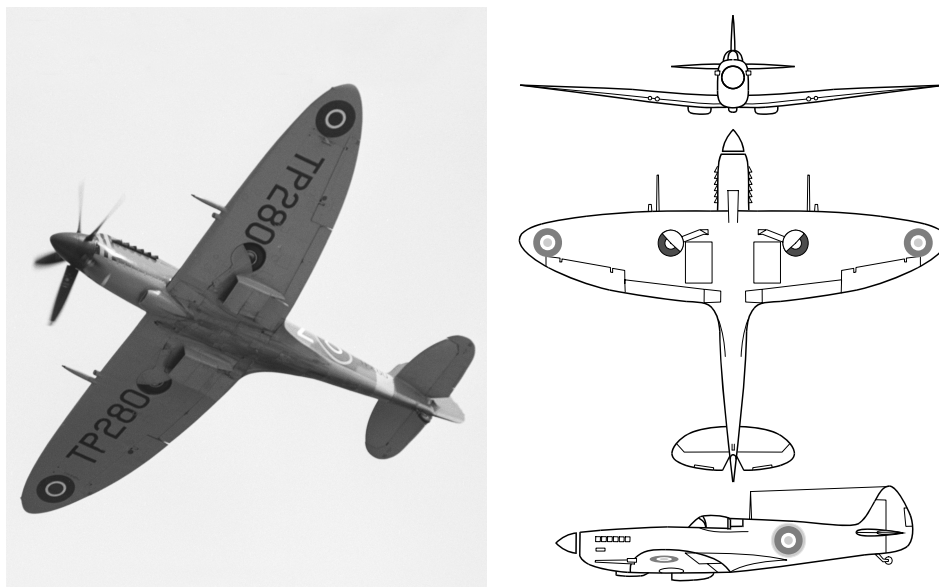


Figure 1.8.11. Elliptic wing used on the famous British Spitfire. (Photograph by Barry Santana)

lift slope for the finite wing approaches that of the airfoil section and the induced drag becomes small. At a given angle of attack, the untwisted elliptic wing produces more lift than any other untwisted wing of the same aspect ratio. Planform shape affects both the induced drag and the lift slope of a finite wing. However, the effect of planform shape is small compared to that of aspect ratio.

While untwisted elliptic wings produce minimum possible induced drag, they are more expensive to manufacture than simple rectangular wings. Untwisted rectangular wings are easy to manufacture, but they generate induced drag at a level that is less than optimum. The untwisted tapered wing, shown in Fig. 1.8.12, has commonly been used as a compromise. Tapered wings have a chord length that varies linearly from the root to the tip. They are nearly as easy to manufacture as rectangular wings, and they can be designed to produce induced drag close to the optimum value of an elliptic wing.

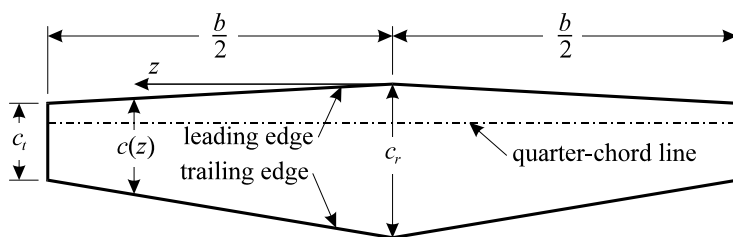


Figure 1.8.12. Planform shape of an unswept tapered wing with a taper ratio of 0.5 and an aspect ratio of 6.

The chord length for a tapered wing varies with the spanwise coordinate according to the relation

$$c(z) = \frac{2b}{R_A(1+R_T)}[1 - (1-R_T)|2z/b|] \quad \text{or} \quad c(\theta) = \frac{2b}{R_A(1+R_T)}[1 - (1-R_T)|\cos \theta|]$$

where R_T is the taper ratio, related to the root chord, c_r , and the tip chord, c_t , by

$$R_T \equiv c_t/c_r$$

For a wing of any planform having no sweep, dihedral, geometric twist, or aerodynamic twist, the circulation distribution predicted from Prandtl's lifting-line theory can be written in terms of a Fourier series with coefficients, a_n , that are independent of angle of attack. Under these conditions, Eq. (1.8.3) can be rearranged to give

$$\Gamma(\theta) = 2bV_\infty(\alpha - \alpha_{L0}) \sum_{n=1}^N a_n \sin(n\theta); \quad \theta = \cos^{-1}(-2z/b) \quad (1.8.9)$$

where the Fourier coefficients are obtained from

$$\sum_{n=1}^N a_n \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = 1 \quad (1.8.10)$$

The lift and induced drag coefficients predicted from Eqs. (1.8.5) and (1.8.6) for a **wing with no geometric or aerodynamic twist** can be written as

$$C_L = C_{L,\alpha}(\alpha - \alpha_{L0}), \quad C_{L,\alpha} = \frac{\tilde{C}_{L,\alpha}}{[1 + \tilde{C}_{L,\alpha}/(\pi R_A)](1 + \kappa_L)} \quad (1.8.11)$$

$$C_{D_i} = \frac{C_L^2}{\pi R_A e_s}, \quad e_s = \frac{1}{1 + \kappa_D} \quad (1.8.12)$$

where the lift slope factor, κ_L , and the induced drag factor, κ_D , are given by

$$\kappa_L = \frac{1 - (1 + \pi R_A / \tilde{C}_{L,\alpha})a_1}{(1 + \pi R_A / \tilde{C}_{L,\alpha})a_1} \quad (1.8.13)$$

$$\kappa_D = \sum_{n=2}^N n \left(\frac{a_n}{a_1} \right)^2 \quad (1.8.14)$$

For untwisted tapered wings, numerical results obtained for these two factors are shown in Figs. 1.8.13 and 1.8.14. These results were generated using a section lift slope of 2π .

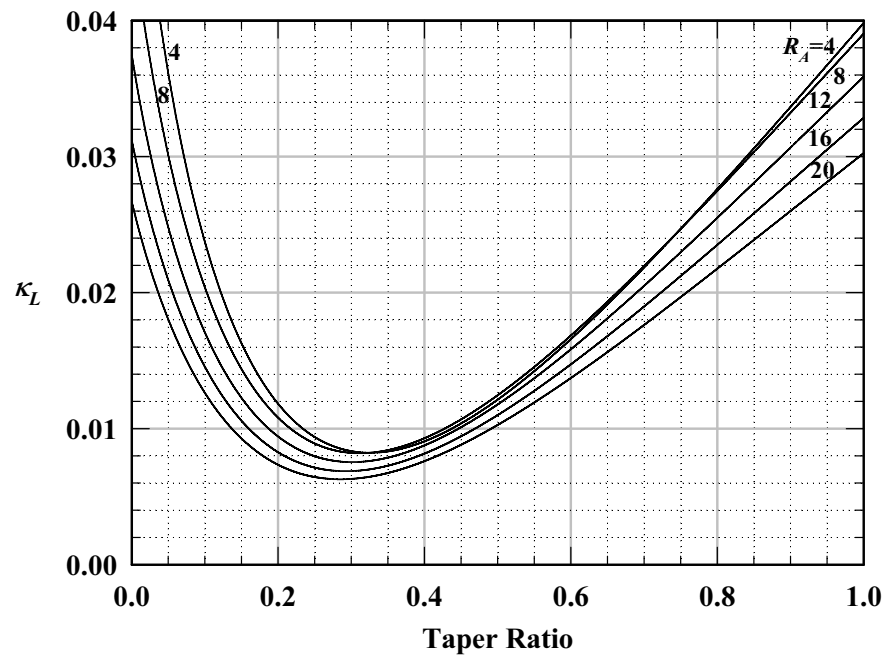


Figure 1.8.13. Lift slope factor for untwisted tapered wings from Prandtl's lifting-line theory.

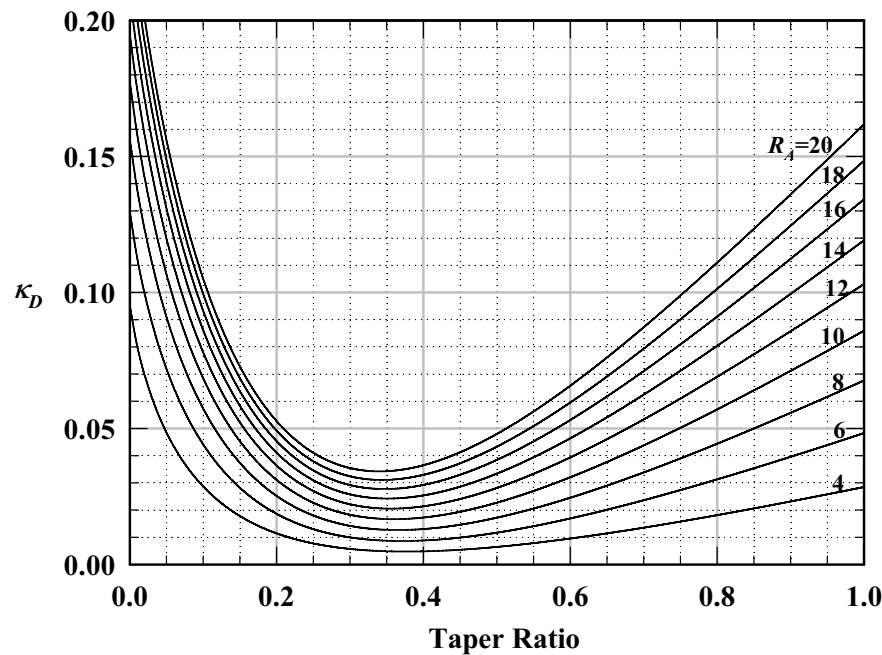


Figure 1.8.14. Induced drag factor for untwisted tapered wings from Prandtl's lifting-line theory.

Glauert (1926) first presented results similar to those shown in Fig. 1.8.14. Such results have sometimes led to the conclusion that a tapered wing with a taper ratio of about 0.4 always produces significantly less induced drag than a rectangular wing of the same aspect ratio developing the same lift. As a result, tapered wings are often used as a means of reducing induced drag. However, this reduction in drag usually comes at a price. Because a tapered wing has a lower Reynolds number at the wingtips than at the root, a tapered wing with no geometric or aerodynamic twist tends to stall first in the region of the wingtips. This wingtip stall commonly leads to poor handling qualities during stall recovery.

The results shown in Fig. 1.8.14 can be misleading if one loses sight of the fact that these results apply only for the special case of wings with no geometric or aerodynamic twist. This is only one of many possible twist distributions that could be used for a wing of any given planform. Furthermore, it is not the twist distribution that produces minimum induced drag with finite lift, except for the case of an elliptic wing. When the effects of twist are included, it can be shown that the conclusions sometimes reached from consideration of only those results shown in Fig. 1.8.14 are erroneous.

Effects of Wing Twist

For a wing with geometric and/or aerodynamic twist, the solution expressed in the form of Eqs. (1.8.3) through (1.8.6) is cumbersome for the evaluation of traditional wing properties, because the Fourier coefficients depend on angle of attack and must be reevaluated for each operating point studied. Furthermore, the definition of σ that is given in Eq. (1.8.6) is not practical for use at an arbitrary angle of attack, because the value becomes singular for a twisted wing when the lift coefficient approaches zero. A more useful form of the solution can be obtained by using the change of variables

$$\alpha(\theta) - \alpha_{L0}(\theta) \equiv (\alpha - \alpha_{L0})_{\text{root}} - \Omega \omega(\theta) \quad (1.8.15)$$

where Ω is defined to be the maximum total washout, geometric plus aerodynamic,

$$\Omega \equiv (\alpha - \alpha_{L0})_{\text{root}} - (\alpha - \alpha_{L0})_{\text{max}} \quad (1.8.16)$$

and $\omega(\theta)$ is the local washout distribution function, which is normalized with respect to maximum total washout

$$\omega(\theta) \equiv \frac{\alpha(\theta) - \alpha_{L0}(\theta) - (\alpha - \alpha_{L0})_{\text{root}}}{(\alpha - \alpha_{L0})_{\text{max}} - (\alpha - \alpha_{L0})_{\text{root}}} \quad (1.8.17)$$

The normalized washout distribution function, $\omega(\theta)$, is independent of angle of attack and always varies from 0.0 at the root to 1.0 at the point of maximum washout, which is commonly at the wingtips.

Using Eq. (1.8.15) in Eq. (1.8.4) gives

$$\sum_{n=1}^N A_n \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = (\alpha - \alpha_{L0})_{\text{root}} - \Omega \omega(\theta) \quad (1.8.18)$$

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The Fourier coefficients in Eq. (1.8.18) for a **wing with geometric and/or aerodynamic twist** can be conveniently written as

$$A_n \equiv a_n(\alpha - \alpha_{L0})_{\text{root}} - b_n \Omega \quad (1.8.19)$$

where

$$\sum_{n=1}^N a_n \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = 1 \quad (1.8.20)$$

$$\sum_{n=1}^N b_n \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = \omega(\theta) \quad (1.8.21)$$

Comparing Eq. (1.8.20) with Eq. (1.8.10), we see that the Fourier coefficients defined by Eq. (1.8.20) are those corresponding to the solution for a wing of the same planform shape but with no geometric or aerodynamic twist. The solution to Eq. (1.8.21) can be obtained in a similar manner and is also independent of angle of attack.

Using Eq. (1.8.19) in Eq. (1.8.5), the lift coefficient for a wing with washout can be expressed as

$$C_L = \pi R_A A_1 = \pi R_A [a_1(\alpha - \alpha_{L0})_{\text{root}} - b_1 \Omega] \quad (1.8.22)$$

Using Eq. (1.8.19) in Eq. (1.8.6), the induced drag coefficient is given by

$$C_{D_i} = \pi R_A \sum_{n=1}^N n A_n^2 = \pi R_A [a_1(\alpha - \alpha_{L0})_{\text{root}} - b_1 \Omega]^2 + \pi R_A \sum_{n=2}^N n [a_n(\alpha - \alpha_{L0})_{\text{root}} - b_n \Omega]^2$$

or after using Eq. (1.8.22) to express the first term on the right-hand side in terms of the lift coefficient,

$$C_{D_i} = \frac{C_L^2}{\pi R_A} + \pi R_A \sum_{n=2}^{\infty} n \left[a_n^2 (\alpha - \alpha_{L0})_{\text{root}}^2 - 2a_n b_n (\alpha - \alpha_{L0})_{\text{root}} \Omega + b_n^2 \Omega^2 \right] \quad (1.8.23)$$

Equations (1.8.22) and (1.8.23) can be algebraically rearranged to yield a convenient expression for the lift and induced drag developed by a **finite wing with geometric and/or aerodynamic twist**:

$$C_L = C_{L,\alpha} [(\alpha - \alpha_{L0})_{\text{root}} - \varepsilon_{\Omega} \Omega] \quad (1.8.24)$$

$$C_{D_i} = \frac{C_L^2 (1 + \kappa_D) - \kappa_{DL} C_L C_{L,\alpha} \Omega + \kappa_{D\Omega} (C_{L,\alpha} \Omega)^2}{\pi R_A} \quad (1.8.25)$$

where

$$C_{L,\alpha} = \pi R_A a_1 = \frac{\tilde{C}_{L,\alpha}}{[1 + \tilde{C}_{L,\alpha}/(\pi R_A)](1 + \kappa_L)} \quad (1.8.26)$$

$$\kappa_L \equiv \frac{1 - (1 + \pi R_A/\tilde{C}_{L,\alpha})a_1}{(1 + \pi R_A/\tilde{C}_{L,\alpha})a_1} \quad (1.8.27)$$

$$\varepsilon_\Omega \equiv \frac{b_1}{a_1} \quad (1.8.28)$$

$$\kappa_D \equiv \sum_{n=2}^N n \frac{a_n^2}{a_1^2} \quad (1.8.29)$$

$$\kappa_{DL} \equiv 2 \frac{b_1}{a_1} \sum_{n=2}^N n \frac{a_n}{a_1} \left(\frac{b_n}{b_1} - \frac{a_n}{a_1} \right) \quad (1.8.30)$$

$$\kappa_{D\Omega} \equiv \left(\frac{b_1}{a_1} \right)^2 \sum_{n=2}^N n \left(\frac{b_n}{b_1} - \frac{a_n}{a_1} \right)^2 \quad (1.8.31)$$

Comparing Eqs. (1.8.24) through (1.8.31) with Eqs. (1.8.11) and (1.8.14), we see that washout increases the zero-lift angle of attack for any wing but the lift slope for a wing of arbitrary planform shape is not affected by washout. Notice that the induced drag for a wing with washout is not zero at zero lift. In addition to the usual component of induced drag, which is proportional to the lift coefficient squared, a wing with washout produces a component of induced drag that is proportional to the washout squared, and this results in induced drag at zero lift. There is also a component of induced drag that varies with the product of the lift coefficient and the washout.

A commonly used washout distribution that is easy to implement is linear washout. For the special case of a linear variation in washout from the root to the tip, the normalized washout distribution function is simply

$$\omega(z) = |2z/b| \quad \text{or} \quad \omega(\theta) = |\cos(\theta)| \quad (1.8.32)$$

For tapered wings, the variation in chord length is also linear:

$$c(z) = \frac{2b}{R_A(1+R_T)} [1 - (1-R_T)|2z/b|] \quad \text{or} \quad c(\theta) = \frac{2b}{R_A(1+R_T)} [1 - (1-R_T)|\cos(\theta)|] \quad (1.8.33)$$

Using Eqs. (1.8.32) and (1.8.33) in Eqs. (1.8.20) and (1.8.21) yields the results for a tapered wing with linear washout,

$$\sum_{n=1}^N a_n \left[\frac{2R_A(1+R_T)}{\tilde{C}_{L,\alpha} [1-(1-R_T)|\cos(\theta)|]} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = 1 \quad (1.8.34)$$

$$\sum_{n=1}^N b_n \left[\frac{2R_A(1+R_T)}{\tilde{C}_{L,\alpha} [1-(1-R_T)|\cos(\theta)|]} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = |\cos(\theta)| \quad (1.8.35)$$

The solution obtained from Eq. (1.8.34) for the Fourier coefficients, a_n , is the familiar result that was used to produce Figs. 1.8.13 and 1.8.14. Induced drag for a wing with washout is readily predicted from Eq. (1.8.25) with the definitions given in Eqs. (1.8.29) through (1.8.31). For tapered wings with linear washout, the Fourier coefficients, b_n , can be obtained from Eq. (1.8.35) in exactly the same manner as the coefficients, a_n , are obtained from Eq. (1.8.34). Using the Fourier coefficients so obtained in Eqs. (1.8.28), (1.8.30), and (1.8.31) produces the results shown in Figs. 1.8.15 through 1.8.17. Notice from examining either Eq. (1.8.31) or Fig. 1.8.17 that $\kappa_{D\Omega}$ is always positive. Thus, the third term in the numerator on the right-hand side of Eq. (1.8.25) always contributes to an increase in induced drag. However, from the results shown in Fig. 1.8.16, we see that the second term in the numerator on the right-hand side of Eq. (1.8.25) can either increase or decrease the induced drag, depending on the signs of κ_{DL} and Ω . This raises an important question regarding wing efficiency. What level and distribution of washout will result in minimum induced drag for a given wing planform and lift coefficient?

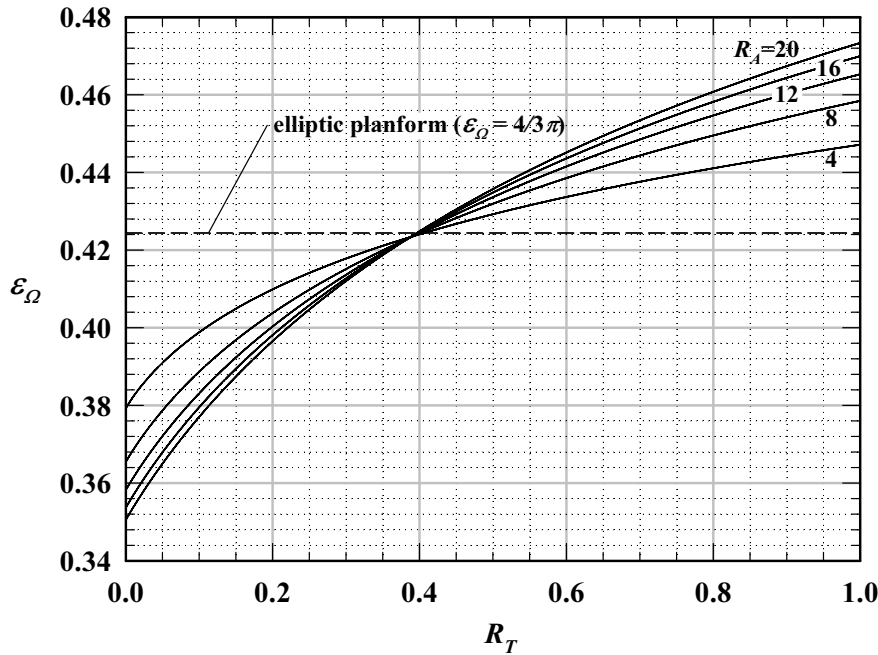


Figure 1.8.15. Washout effectiveness for tapered wings with linear washout.

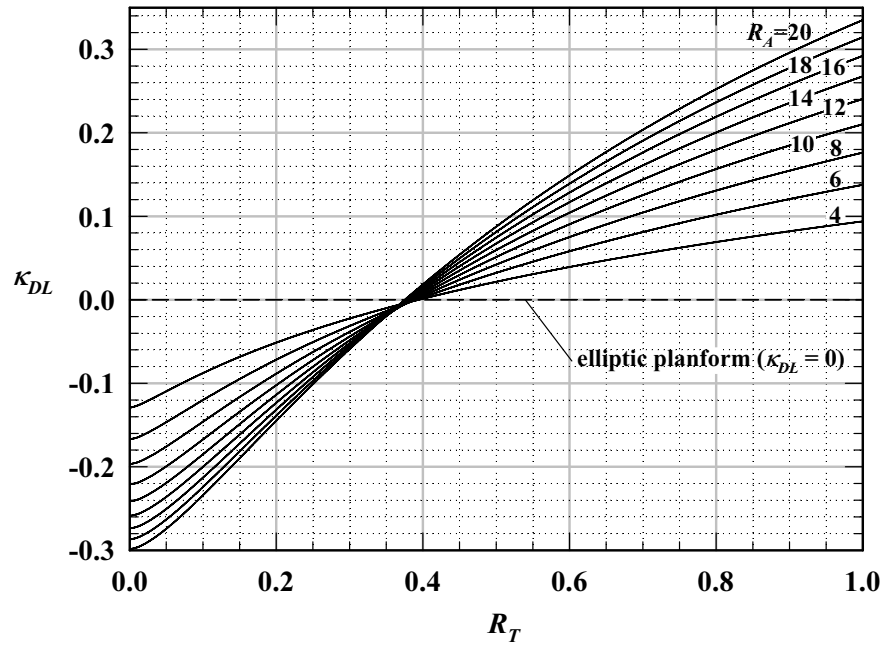


Figure 1.8.16. Lift-washout contribution to induced drag for tapered wings with linear washout.

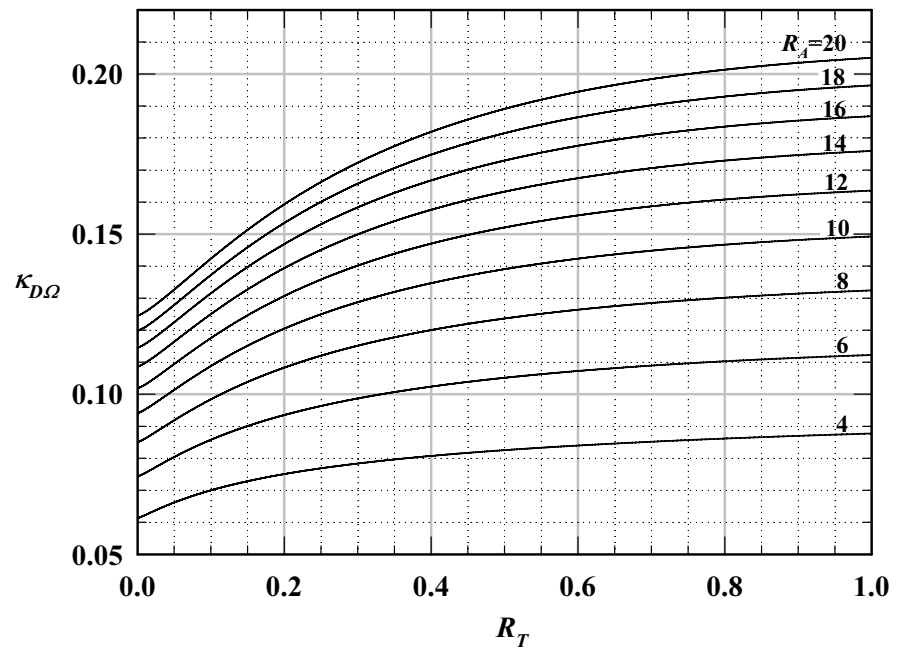


Figure 1.8.17. Washout contribution to induced drag for tapered wings with linear washout.

Minimizing Induced Drag with Washout

For a wing of any given planform shape having a fixed washout distribution, induced drag can be minimized with washout as a result of the trade-off between the second and third terms in the numerator on the right-hand side of Eq. (1.8.25). Thus, Eq. (1.8.25) can be used to determine the optimum value of total washout, which will result in minimum induced drag for any washout distribution function and any specified lift coefficient. Differentiating Eq. (1.8.25) with respect to Ω at constant lift coefficient gives

$$\frac{\partial C_{Di}}{\partial \Omega} = \frac{-\kappa_{DL} C_L C_{L,\alpha} + 2\kappa_{D\Omega} C_{L,\alpha}^2 \Omega}{\pi R_A} \quad (1.8.36)$$

Setting the right-hand side of Eq. (1.8.36) to zero and solving for Ω , it can be seen that minimum induced drag is attained for any given wing planform, $c(z)$, any given washout distribution, $\omega(z)$, and any given design lift coefficient, C_{Ld} , by using an optimum total amount of washout, Ω_{opt} , which is given by the relation

$$\Omega_{\text{opt}} = \frac{\kappa_{DL} C_{Ld}}{2\kappa_{D\Omega} C_{L,\alpha}} \quad (1.8.37)$$

For the elliptic planform, all of the Fourier coefficients, a_n , are zero for n greater than 1. Thus, Eq. (1.8.30) shows that κ_{DL} is zero for an elliptic wing. As a result, elliptic wings are always optimized with no washout. From consideration of Eq. (1.8.37) together with the results shown in Figs. 1.8.16 and 1.8.17, we see that tapered wings with linear washout and taper ratios greater than 0.4 are optimized with positive washout, whereas those with taper ratios less than about 0.4 are optimized with negative washout.

Using the value of optimum washout from Eq. (1.8.37) in the expression for induced drag coefficient given by Eq. (1.8.25), we find that the induced drag coefficient for a wing of arbitrary planform with a fixed washout distribution and optimum total washout is given by

$$(C_{Di})_{\text{opt}} = \frac{C_L^2}{\pi R_A} \left[1 + \kappa_D - \frac{\kappa_{DL}^2}{4\kappa_{D\Omega}} \left(2 - \frac{C_{Ld}}{C_L} \right) \frac{C_{Ld}}{C_L} \right] \quad (1.8.38)$$

From Eq. (1.8.38) it can be seen that a wing with optimum washout will always produce less induced drag than a wing with no washout having the same planform and aspect ratio, provided that the actual lift coefficient is greater than one-half the design lift coefficient. When the actual lift coefficient is equal to the design lift coefficient, the induced drag coefficient for a wing with optimum washout is

$$(C_{Di})_{\text{opt}} = \frac{C_L^2}{\pi R_A} (1 + \kappa_{Do}), \quad \kappa_{Do} \equiv \kappa_D - \frac{\kappa_{DL}^2}{4\kappa_{D\Omega}} \quad (1.8.39)$$

For tapered wings with linear washout, the variations in κ_{Do} with aspect ratio and taper ratio are shown in Fig. 1.8.18. For comparison, the dashed lines in this figure show the

same results for wings with no washout. Notice that when linear washout is used to further optimize tapered wings, taper ratios near 0.4 correspond closely to a maximum in induced drag, not to a minimum.

The choice of a linear washout distribution, which was used to generate the results shown in Fig. 1.8.18, is as arbitrary as the choice of no washout. While a linear variation in washout is commonly used and simple to implement, it is not the optimum washout distribution for wings with linear taper. Minimum possible induced drag for a finite lift coefficient always occurs when the local section lift varies with the spanwise coordinate in proportion to $\sin(\theta)$. This results in uniform downwash and requires that the product of the local chord length and local aerodynamic angle of attack, $\alpha - \alpha_{L0}$, varies elliptically with the spanwise coordinate, i.e.,

$$\frac{c(z)[\alpha(z) - \alpha_{L0}(z)]}{\sqrt{1 - (2z/b)^2}} = \frac{c(\theta)[\alpha(\theta) - \alpha_{L0}(\theta)]}{\sin(\theta)} = \text{constant} \quad (1.8.40)$$

There are many possibilities for wing geometry that will satisfy this condition. The elliptic planform with no geometric or aerodynamic twist is only one such geometry. Since the local aerodynamic angle of attack decreases along the span in direct proportion to the increase in washout, Eq. (1.8.40) can only be satisfied if the washout distribution satisfies the relation

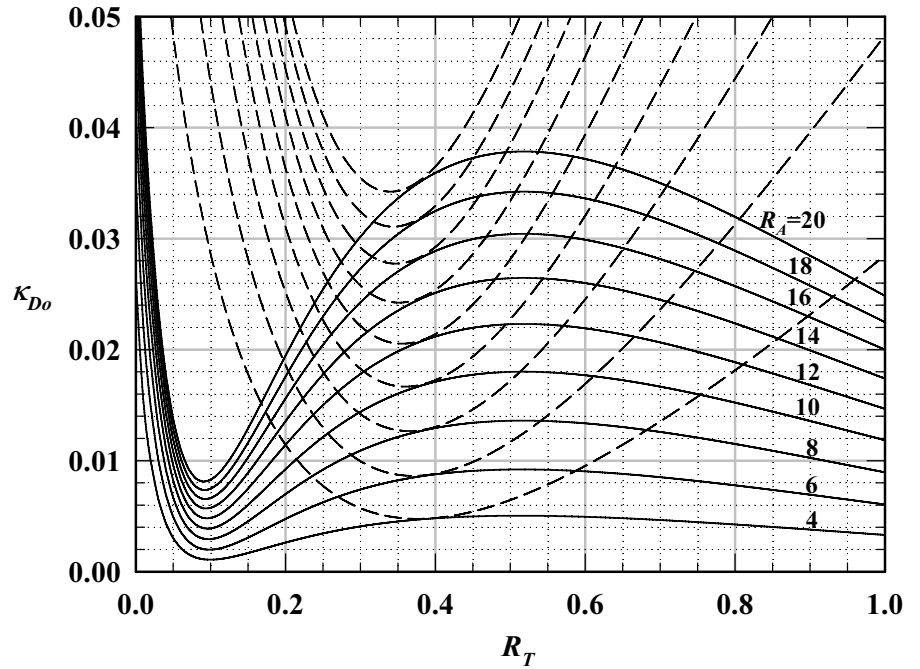


Figure 1.8.18. Optimum induced drag factor for tapered wings with linear washout.

$$\frac{c(z)[1-\omega(z)]}{\sqrt{1-(2z/b)^2}} = \frac{c(\theta)[1-\omega(\theta)]}{\sin(\theta)} = \text{constant} \quad (1.8.41)$$

Equation (1.8.41) is satisfied by the **optimum washout distribution**

$$\omega_{\text{opt}} = 1 - \frac{\sqrt{1-(2z/b)^2}}{c(z)/c_{\text{root}}} = 1 - \frac{\sin(\theta)}{c(\theta)/c_{\text{root}}}, \quad \Omega_{\text{opt}} = \frac{4b C_L}{\pi R_A \tilde{C}_{L,\alpha} c_{\text{root}}} \quad (1.8.42)$$

For wings with linear taper, this gives

$$\omega_{\text{opt}} = 1 - \frac{\sqrt{1-(2z/b)^2}}{1-(1-R_T)|2z/b|} = 1 - \frac{\sin(\theta)}{1-(1-R_T)|\cos(\theta)|}, \quad \Omega_{\text{opt}} = \frac{2(1+R_T)C_L}{\pi \tilde{C}_{L,\alpha}} \quad (1.8.43)$$

This optimum washout distribution is shown in Fig. 1.8.19 for several values of taper ratio. Results obtained for tapered wings having this washout distribution are presented in Figs. 1.8.20 through 1.8.22. It should be noted that computing Ω_{opt} from Eq. (1.8.42) or (1.8.43) is only valid for wings having the optimum washout distribution, ω_{opt} . On the other hand, Eq. (1.8.37) is valid for any washout distribution. For $\omega = \omega_{\text{opt}}$, Eq. (1.8.37) produces exactly the same result as Eq. (1.8.42) or (1.8.43).

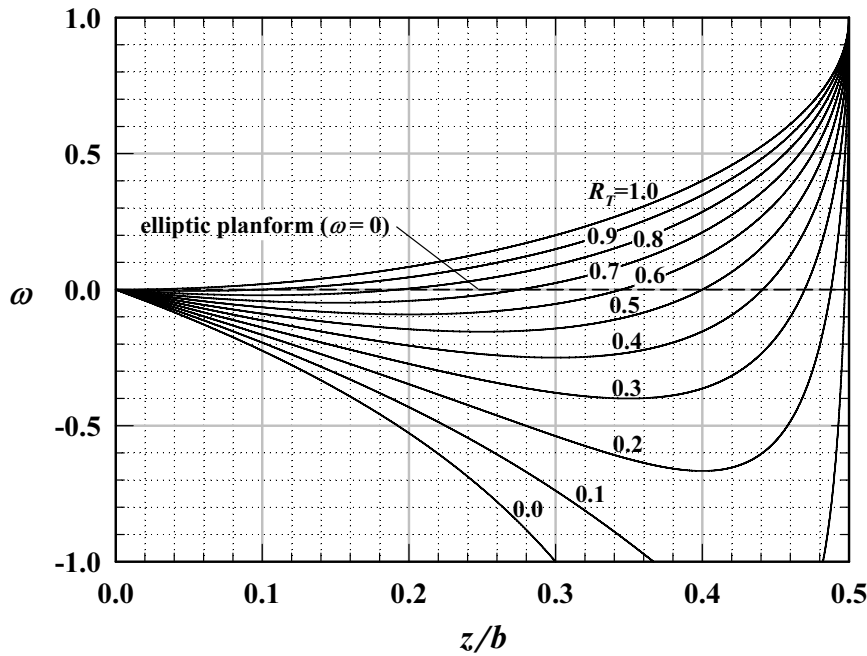


Figure 1.8.19. Optimum washout distribution that results in production of minimum induced drag for wings with linear taper, as defined in Eq. (1.8.43).

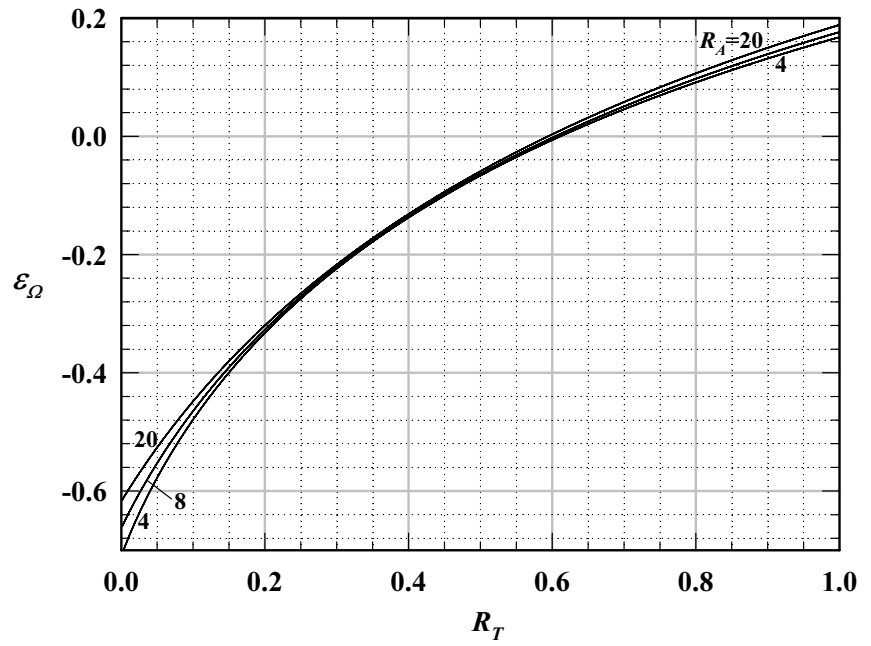


Figure 1.8.20. Washout effectiveness for wings with linear taper and optimum washout.

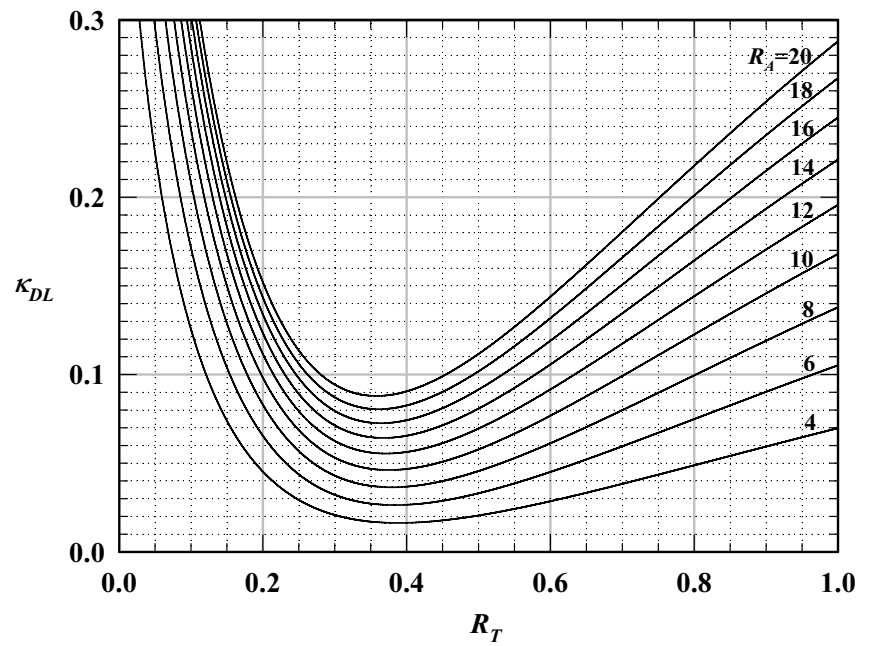


Figure 1.8.21. Lift-washout contribution to the induced drag factor for wings with linear taper and the optimum washout distribution specified by Eq. (1.8.43).

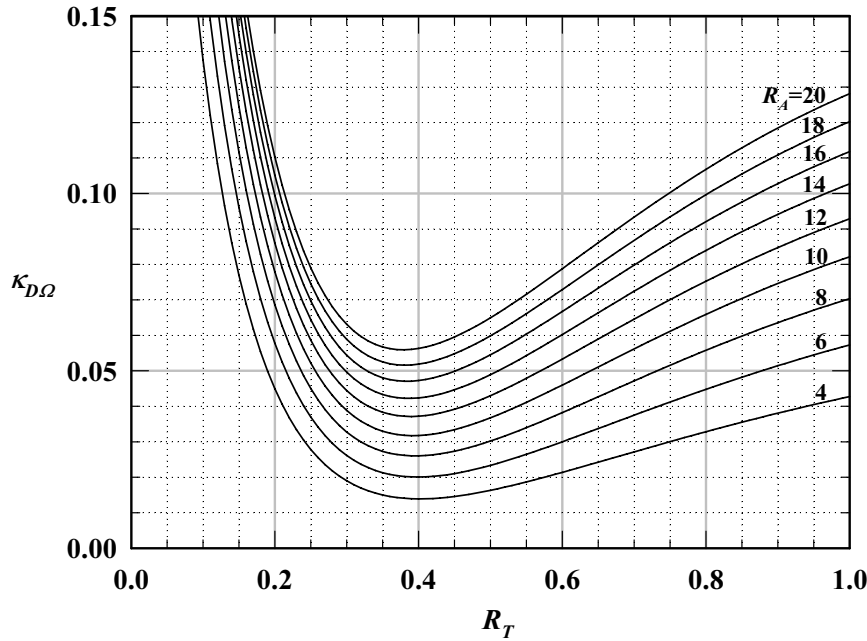


Figure 1.8.22. Washout contribution to the induced drag factor for wings with linear taper and the optimum washout distribution specified by Eq. (1.8.43).

When an unswept wing of arbitrary planform has the washout distribution specified by Eq. (1.8.42), the value of $\kappa_{D\omega}$ as defined in Eq. (1.8.39) is always identically zero. With the washout distribution set to ω_{opt} and the total amount washout set to Ω_{opt} , an unswept wing of any planform shape can be designed to operate at a given lift coefficient with the same induced drag as that produced by an untwisted elliptic wing with the same aspect ratio and lift coefficient. This is demonstrated in the following examples.

EXAMPLE 1.8.1. When results similar to those shown in Fig. 1.8.14 were first published by Glauert (1926), all calculations were made without the aid of a computer. For this reason, only a few terms were retained in the Fourier series. For a rectangular planform with an aspect ratio of 8.0 and the optimum washout distribution specified by Eq. (1.8.42), solve Eqs. (1.8.20) and (1.8.21) by retaining only seven terms in the Fourier series and forcing Eqs. (1.8.20) and (1.8.21) to be satisfied at only seven sections along the span of the wing. Locate the first and last sections at the wingtips and space the intermediate sections equally in θ , i.e.,

$$\theta_i = \frac{(i-1)\pi}{N-1} \quad i = 1, N$$

where N is 7. Assume an airfoil lift slope of 2π and use the solution to compute κ_D , κ_{DL} , $\kappa_{D\Omega}$ and $\kappa_{D\omega}$. Compare the results with those obtained for $N=99$.

Solution. In order to obtain N independent equations for the N unknown Fourier coefficients, a_n , Eq. (1.8.20) can be written for each of N spanwise sections of the wing. With the first and last sections located at the wingtips and the intermediate sections spaced equally in θ , this gives the system of equations

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & \cdots & C_{1N} \\ C_{21} & C_{22} & C_{23} & \cdots & C_{2N} \\ C_{31} & C_{32} & C_{33} & \cdots & C_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C_{N1} & C_{N2} & C_{N3} & \cdots & C_{NN} \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_N \end{Bmatrix} = \begin{Bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{Bmatrix}$$

where

$$C_{ij} = \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta_i)} + \frac{j}{\sin(\theta_i)} \right] \sin(j\theta_i), \quad \theta_i = \frac{(i-1)\pi}{N-1}, \quad \begin{cases} i=1, N \\ j=1, N \end{cases}$$

The components of the matrix, $[\mathbf{C}]$, on the left-hand side of this linear system are indeterminate when evaluated at the wingtips, $\theta=0$ and $\theta=\pi$. However, applying *l'Hôpital's rule* gives

$$\begin{aligned} \left(\frac{\sin(j\theta_i)}{\sin(\theta_i)} \right)_{\theta_i \rightarrow 0} &= \left(\frac{j \cos(j\theta_i)}{\cos(\theta_i)} \right)_{\theta_i \rightarrow 0} = j \\ \left(\frac{\sin(j\theta_i)}{\sin(\theta_i)} \right)_{\theta_i \rightarrow \pi} &= \left(\frac{j \cos(j\theta_i)}{\cos(\theta_i)} \right)_{\theta_i \rightarrow \pi} = (-1)^{j+1} j \end{aligned}$$

Thus, for the purpose of numerical evaluation, the matrix $[\mathbf{C}]$ is determined from the algorithm

$$\begin{aligned} C_{1j} &= j^2, \quad j=1, N \\ C_{ij} &= \left[\frac{4b}{\tilde{C}_{L,\alpha} c(\theta_i)} + \frac{j}{\sin(\theta_i)} \right] \sin(j\theta_i), \quad \theta_i = \frac{(i-1)\pi}{N-1}, \quad \begin{cases} i=2, N-1 \\ j=1, N \end{cases} \\ C_{Nj} &= (-1)^{j+1} j^2, \quad j=1, N \end{aligned}$$

For this rectangular planform of aspect ratio of 8.0, the chord is independent of θ and $b/c(\theta)=8$. Thus, for an airfoil section lift slope of 2π and $N=7$, we have

$$C_{ij} = \left[\frac{16}{\pi} + \frac{j}{\sin[(i-1)\pi/6]} \right] \sin[j(i-1)\pi/6], \quad \begin{cases} i=2, 6 \\ j=1, 7 \end{cases}$$

Some example computations give

$$C_{21} = \left[\frac{16}{\pi} + \frac{1}{\sin[(2-1)\pi/6]} \right] \sin\left(\frac{1(2-1)\pi}{6}\right) = \left[\frac{16}{\pi} + \frac{1}{\sin(\pi/6)} \right] \sin\left(\frac{\pi}{6}\right) = 3.5465$$

$$C_{22} = \left[\frac{16}{\pi} + \frac{2}{\sin[(2-1)\pi/6]} \right] \sin\left(\frac{2(2-1)\pi}{6}\right) = \left[\frac{16}{\pi} + \frac{2}{\sin(\pi/6)} \right] \sin\left(\frac{2\pi}{6}\right) = 7.8747$$

$$C_{45} = \left[\frac{16}{\pi} + \frac{5}{\sin[(4-1)\pi/6]} \right] \sin\left(\frac{5(4-1)\pi}{6}\right) = \left[\frac{16}{\pi} + \frac{5}{\sin(3\pi/6)} \right] \sin\left(\frac{15\pi}{6}\right) = 10.093$$

After performing the remaining computations, we have

$$[C] = \begin{bmatrix} 1 & 4 & 9 & 16 & 25 & 36 & 49 \\ 3.5465 & 7.8747 & 11.093 & 11.339 & 7.5465 & 0 & -9.5465 \\ 5.4106 & 6.4106 & 0 & -8.4106 & -9.4106 & 0 & 11.411 \\ 6.0930 & 0 & -8.0930 & 0 & 10.093 & 0 & -12.093 \\ 5.4106 & -6.4106 & 0 & 8.4106 & -9.4106 & 0 & 11.411 \\ 3.5465 & -7.8747 & 11.093 & -11.339 & 7.5465 & 0 & -9.5465 \\ 1 & -4 & 9 & -16 & 25 & -36 & 49 \end{bmatrix}$$

The inverse of this matrix is

$$[C]^{-1} = \begin{bmatrix} 0.00005 & 0.02061 & 0.04702 & 0.05661 & 0.04702 & 0.02061 & 0.00005 \\ 0.00000 & 0.03027 & 0.04081 & 0.00000 & -0.04081 & -0.03027 & 0.00000 \\ 0.00019 & 0.03016 & 0.00298 & -0.04047 & 0.00298 & 0.03016 & 0.00019 \\ 0.00000 & 0.02307 & -0.02834 & 0.00000 & 0.02834 & -0.02307 & 0.00000 \\ 0.00764 & 0.00286 & -0.01725 & 0.02481 & -0.01725 & 0.00286 & 0.00764 \\ 0.01389 & -0.01362 & 0.00806 & 0.00000 & -0.00806 & 0.01362 & -0.01389 \\ 0.00627 & -0.00742 & 0.00729 & -0.00638 & 0.00729 & -0.00742 & 0.00627 \end{bmatrix}$$

which yields

$$\begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{Bmatrix} = [C]^{-1} \begin{Bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 0.191966 \\ 0 \\ 0.026191 \\ 0 \\ 0.011287 \\ 0 \\ 0.005921 \end{Bmatrix}$$

Comparing Eqs. (1.8.20) and (1.8.21), we see that the Fourier coefficients, b_n , can be obtained from a linear system of equations that is almost identical to that used

to obtain the Fourier coefficients, a_n . The matrices on the left-hand sides of these two systems are identical. The only difference is that the right-hand side of the system used to obtain a_n is a vector having all components equal to 1, whereas the right-hand side of the system used to obtain b_n is a vector obtained from the washout distribution function,

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & \cdots & C_{1N} \\ C_{21} & C_{22} & C_{23} & \cdots & C_{2N} \\ C_{31} & C_{32} & C_{33} & \cdots & C_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C_{N1} & C_{N2} & C_{N3} & \cdots & C_{NN} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_N \end{bmatrix} = \begin{bmatrix} \omega(\theta_1) \\ \omega(\theta_2) \\ \omega(\theta_3) \\ \vdots \\ \omega(\theta_N) \end{bmatrix}$$

Using the optimum washout distribution function from Eq. (1.8.42) yields

$$\omega(\theta_i) = 1 - \frac{\sin(\theta_i)}{c(\theta_i)/c_{\text{root}}}, \quad \theta_i = \frac{(i-1)\pi}{N-1}, \quad i=1, N$$

For this rectangular planform, the chord is constant, so $c(\theta_i)/c_{\text{root}}=1$. Thus, for $N=7$, the Fourier coefficients, b_n , are found to be

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \\ b_7 \end{bmatrix} = [\mathbf{C}]^{-1} \begin{bmatrix} 1 \\ 1 - \sin(\pi/6) \\ 1 - \sin(2\pi/6) \\ 1 - \sin(3\pi/6) \\ 1 - \sin(4\pi/6) \\ 1 - \sin(5\pi/6) \\ 1 - \sin(6\pi/6) \end{bmatrix} = [\mathbf{C}]^{-1} \begin{bmatrix} 1 \\ 0.5 \\ 0.133975 \\ 0 \\ 0.133975 \\ 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.033309 \\ 0 \\ 0.031334 \\ 0 \\ 0.013504 \\ 0 \\ 0.007084 \end{bmatrix}$$

Using these results in Eqs. (1.8.29), (1.8.30), (1.8.31), and (1.8.39) yields

$$\kappa_D \equiv \sum_{n=2}^N n \frac{a_n^2}{a_1^2} = \underline{0.079791}$$

$$\kappa_{DL} \equiv 2 \frac{b_1}{a_1} \sum_{n=2}^N n \frac{a_n}{a_1} \left(\frac{b_n}{b_1} - \frac{a_n}{a_1} \right) = \underline{0.163225}$$

$$\kappa_{D\Omega} \equiv \left(\frac{b_1}{a_1} \right)^2 \sum_{n=2}^N n \left(\frac{b_n}{b_1} - \frac{a_n}{a_1} \right)^2 = \underline{0.083476}$$

$$\kappa_{Do} \equiv \kappa_D - \frac{\kappa_{DL}^2}{4\kappa_{D\Omega}} = \underline{0.000000}$$

For comparison, repeating these computations with $N=99$ gives

$$\begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \\ \vdots \\ a_{98} \\ a_{99} \end{Bmatrix} = \begin{Bmatrix} 0.19248612 \\ 0 \\ 0.02740767 \\ 0 \\ 0.00656477 \\ 0 \\ 0.00202851 \\ \vdots \\ 0 \\ 0.00000144 \end{Bmatrix}, \quad \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \\ b_7 \\ \vdots \\ b_{98} \\ b_{99} \end{Bmatrix} = \begin{Bmatrix} 0.03393114 \\ 0 \\ 0.03278916 \\ 0 \\ 0.00785376 \\ 0 \\ 0.00242681 \\ \vdots \\ 0 \\ 0.00000172 \end{Bmatrix}$$

$$\kappa_D = \underline{0.067611}, \quad \kappa_{DL} = \underline{0.137937}, \quad \kappa_{D\Omega} = \underline{0.070353}, \quad \kappa_{Do} = \underline{0.000000}$$

EXAMPLE 1.8.2. We wish to repeat the computations in Example 1.8.1 for a wing of tapered planform with an aspect ratio of 8.0 and a taper ratio of 0.5.

Solution. For a wing planform having linear taper, the section chord length varies with the spanwise coordinate as specified in Eq. (1.8.33). With an aspect ratio of 8.0 and a taper ratio of 0.5, this gives

$$\frac{c(\theta)}{b} = \frac{2}{R_A(1+R_T)} [1 - (1-R_T)|\cos(\theta)|] = \frac{1}{6} [1 - 0.5|\cos(\theta)|]$$

Thus, following Example 1.8.1, for this wing geometry and an airfoil section lift slope of 2π , the components of the matrix $[\mathbf{C}]$ are determined from

$$\begin{aligned} C_{1j} &= j^2, \quad j = 1, N \\ C_{ij} &= \left[\frac{12}{\pi[1 - 0.5|\cos(\theta_i)|]} + \frac{j}{\sin(\theta_i)} \right] \sin(j\theta_i), \quad \theta_i = \frac{(i-1)\pi}{N-1}, \quad \begin{cases} i = 2, N-1 \\ j = 1, N \end{cases} \\ C_{Nj} &= (-1)^{j+1} j^2, \quad j = 1, N \end{aligned}$$

For example, with $N=7$,

$$C_{45} = \left[\frac{12}{\pi[1 - 0.5|\cos(3\pi/6)|]} + \frac{5}{\sin(3\pi/6)} \right] \sin(15\pi/6) = 8.8197$$

After performing the remaining computations, we have

$$[\mathbf{C}] = \begin{bmatrix} 1 & 4 & 9 & 16 & 25 & 36 & 49 \\ 4.3684 & 9.2984 & 12.7369 & 12.7625 & 8.3684 & 0 & -10.3684 \\ 5.4106 & 6.4106 & 0 & -8.4106 & -9.4106 & 0 & 11.4106 \\ 4.8197 & 0 & -6.8197 & 0 & 8.8197 & 0 & -10.8197 \\ 5.4106 & -6.4106 & 0 & 8.4106 & -9.4106 & 0 & 11.4106 \\ 4.3684 & -9.2984 & 12.7369 & -12.7625 & 8.3684 & 0 & -10.3684 \\ 1 & -4 & 9 & -16 & 25 & -36 & 49 \end{bmatrix}$$

$$[\mathbf{C}]^{-1} =$$

$$\begin{bmatrix} 0.00010 & 0.01798 & 0.04785 & 0.06742 & 0.04785 & 0.01798 & 0.00010 \\ 0.00000 & 0.02628 & 0.03988 & 0.00000 & -0.03988 & -0.02628 & 0.00000 \\ 0.00003 & 0.02619 & 0.00055 & -0.04872 & 0.00055 & 0.02619 & 0.00003 \\ 0.00000 & 0.02003 & -0.02905 & 0.00000 & 0.02905 & -0.02003 & 0.00000 \\ 0.00767 & 0.00251 & -0.01663 & 0.02962 & -0.01663 & 0.00251 & 0.00767 \\ 0.01389 & -0.01182 & 0.00848 & 0.00000 & -0.00848 & 0.01182 & -0.01389 \\ 0.00628 & -0.00646 & 0.00741 & -0.00754 & 0.00741 & -0.00646 & 0.00628 \end{bmatrix}$$

$$\begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{Bmatrix} = [\mathbf{C}]^{-1} \begin{Bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{Bmatrix} = \begin{Bmatrix} 0.199278 \\ 0 \\ 0.004824 \\ 0 \\ 0.016713 \\ 0 \\ 0.006928 \end{Bmatrix}$$

For this planform with linear taper and a taper ratio of 0.5, the optimum washout distribution function from Eq. (1.8.43) is

$$\omega(\theta_i) = 1 - \frac{\sin(\theta_i)}{1 - 0.5|\cos(\theta_i)|}, \quad \theta_i = \frac{(i-1)\pi}{N-1}, \quad i = 1, N$$

$$\begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \\ b_7 \end{Bmatrix} = [\mathbf{C}]^{-1} \begin{Bmatrix} \omega(0) \\ \omega(\pi/6) \\ \omega(2\pi/6) \\ \omega(3\pi/6) \\ \omega(4\pi/6) \\ \omega(5\pi/6) \\ \omega(\pi) \end{Bmatrix} = [\mathbf{C}]^{-1} \begin{Bmatrix} 1 \\ 0.118146 \\ -0.154701 \\ 0 \\ -0.154701 \\ 0.118146 \\ 1 \end{Bmatrix} = \begin{Bmatrix} -0.010351 \\ 0 \\ 0.006087 \\ 0 \\ 0.021088 \\ 0 \\ 0.008742 \end{Bmatrix}$$

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Using these results in Eqs. (1.8.29), (1.8.30), (1.8.31), and (1.8.39) yields

$$\kappa_D = \underline{0.045387}, \quad \kappa_{DL} = \underline{0.119253}, \quad \kappa_{D\Omega} = \underline{0.078334}, \quad \kappa_{Do} = \underline{0.000000}$$

With $N = 99$ we obtain

$$\begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \\ \vdots \\ a_{98} \\ a_{99} \end{Bmatrix} = \begin{Bmatrix} 0.19751337 \\ 0 \\ 0.00837113 \\ 0 \\ 0.00918923 \\ 0 \\ 0.00142163 \\ \vdots \\ 0 \\ 0.00000164 \end{Bmatrix}, \quad \begin{Bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \\ b_5 \\ b_6 \\ b_7 \\ \vdots \\ b_{98} \\ b_{99} \end{Bmatrix} = \begin{Bmatrix} -0.01257714 \\ 0 \\ 0.01056269 \\ 0 \\ 0.01159497 \\ 0 \\ 0.00179381 \\ \vdots \\ 0 \\ 0.00000207 \end{Bmatrix}$$

$$\kappa_D = \underline{0.017190}, \quad \kappa_{DL} = \underline{0.045569}, \quad \kappa_{D\Omega} = \underline{0.030200}, \quad \kappa_{Do} = \underline{0.000000}$$

From Examples 1.8.1 and 1.8.2 it should be observed that all of the even Fourier coefficients, in both a_n and b_n , were identically zero for both the rectangular and tapered wings. This was realized regardless of how many terms were carried in the Fourier series and was a direct result of the spanwise symmetry of these two wings. If we were always willing to restrict our analysis to spanwise symmetric wing geometry, the computation time could be reduced by forcing all even coefficients to be zero and solving for the odd coefficients using sections distributed over only one side of the wing, i.e., $0 \leq \theta \leq \pi/2$. The requirement for a spanwise symmetric wing planform is not too restrictive. Most wings exhibit this symmetry. However, restricting our analysis to wings having a spanwise symmetric distribution in geometric and aerodynamic twist would eliminate the possibility of using lifting-line theory for the analysis of asymmetric control surface deflection. Such control surface deflection is almost always used in the wing of an airplane to provide roll control.

Solution with Control Surface Deflection and Rolling Rate

Trailing-edge flaps extending over only some portion of the wingspan are commonly used as control surfaces on an airplane wing. A spanwise symmetric control surface deflection can be used to provide pitch control, and spanwise asymmetric control surface deflection can be used to provide roll control. The control surfaces commonly used to provide roll control are called *ailerons*. These are small trailing-edge flaps located in the outboard sections of the wing. The ailerons are deflected asymmetrically to change the rolling moment. One aileron is deflected downward and the other is deflected upward, as

shown in Fig. 1.8.23. This increases lift on the semispan with the downward-deflected aileron and decreases lift on the semispan with the upward-deflected aileron. Aileron deflection is given the symbol δ_a , and the rolling moment is denoted as ℓ . The sign convention that is commonly used for aileron deflection is shown in Fig. 1.8.23. Aileron deflection is assumed positive when the right aileron is deflected downward and the left aileron is deflected upward. The traditional sign convention used for the rolling moment is positive to the right (i.e., a moment that would roll the right wing down). With these two sign conventions, a positive aileron deflection produces a rolling moment to the left (i.e., negative ℓ). The two ailerons are not necessarily deflected the same magnitude. The aileron angle, δ_a , is usually defined to be the average of the two angular deflections.

The rolling moment produced by aileron deflection results in a rolling acceleration, which in turn produces a rolling rate. This rolling rate changes the spanwise variation in geometric angle of attack as shown in Fig. 1.8.24. The symbol traditionally used to

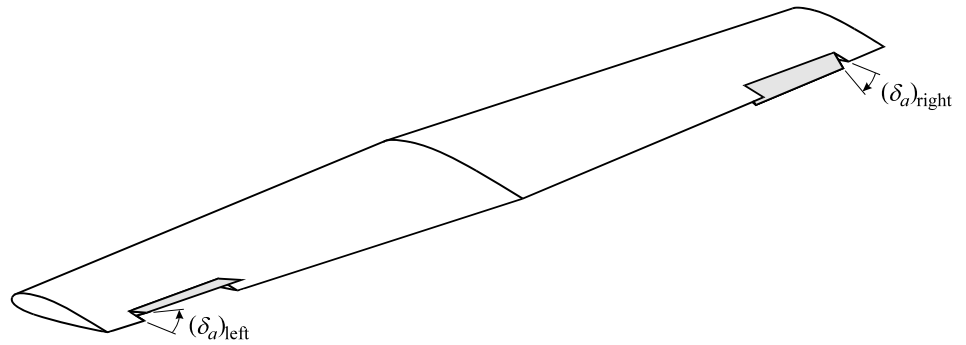


Figure 1.8.23. Asymmetric aileron deflection typically used for roll control.

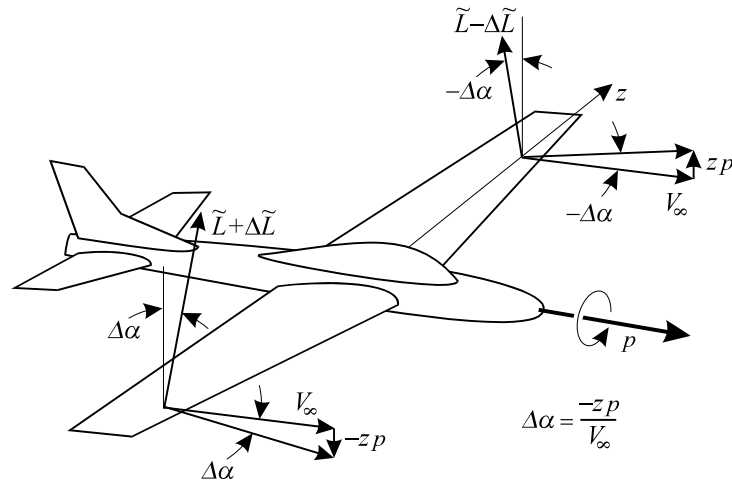


Figure 1.8.24. Change in local section angle of attack resulting from a rolling rate.

represent rolling rate is p . As shown in Fig. 1.8.24, p is taken to be positive when the right semispan is moving downward and the left semispan is moving upward. Thus, a positive rolling rate increases the local geometric angle of attack on the right semispan and decreases the local geometric angle of attack on the left semispan. This produces a negative rolling moment, which opposes the rolling rate.

When a small positive aileron deflection is first initiated, a negative rolling moment is produced by the asymmetric change in the wing's local aerodynamic angle of attack. This rolling moment imbalance results in negative rolling acceleration, which produces a negative rolling rate. As the magnitude of the rolling rate increases, an opposing rolling moment develops as a result of the asymmetric change in the local geometric angle of attack. This reduces the moment imbalance and slows the rolling acceleration. At some point, the positive rolling moment produced by the negative rolling rate will just balance the negative rolling moment produced by the positive aileron deflection, and a constant rolling rate will develop that is proportional to the aileron deflection.

As we will learn in Chapter 9, roll in a conventional airplane is what is referred to as *heavily damped motion*. This means that, when a typical fixed-wing aircraft responds to aileron input, the period of rolling acceleration is very short and the airplane quickly approaches a steady rolling rate, which is proportional to aileron deflection. This gives the pilot the perception that aileron input commands the airplane's rolling rate. Thus, analysis of an airplane's roll response must include consideration of the rolling moment produced by the rolling rate as well as that produced by aileron deflection. Lifting-line theory provides the capability to do just that.

In general, a wing's spanwise variation in local aerodynamic angle of attack can be expressed as the value at the wing root plus the changes due to washout, control surface deflection, and rolling rate,

$$\alpha(z) - \alpha_{L0}(z) \equiv (\alpha - \alpha_{L0})_{\text{root}} - \Omega \omega(z) + \delta_a \chi(z) - pz/V_\infty \quad (1.8.44)$$

where Ω is defined to be the maximum total symmetric washout for the wing, geometric plus aerodynamic,

$$\Omega \equiv [(\alpha - \alpha_{L0})_{\text{root}} - (\Delta\alpha - \Delta\alpha_{L0})_{\text{max}}]_{\text{washout}} \quad (1.8.45)$$

$\omega(z)$ is the symmetric washout distribution function,

$$\omega(z) \equiv \left[\frac{\Delta\alpha(z) - \Delta\alpha_{L0}(z) - (\alpha - \alpha_{L0})_{\text{root}}}{(\Delta\alpha - \Delta\alpha_{L0})_{\text{max}} - (\alpha - \alpha_{L0})_{\text{root}}} \right]_{\text{washout}} \quad (1.8.46)$$

and $\chi(z)$ is the control surface distribution function,

$$\chi(z) \equiv \left[\frac{\Delta\alpha(z) - \Delta\alpha_{L0}(z) - (\alpha - \alpha_{L0})_{\text{root}}}{\delta_a} \right]_{\text{control}} \quad (1.8.47)$$

For example, ailerons extending from the spanwise coordinate z_{ar} to z_{at} give

$$\chi(z) \equiv \begin{cases} 0, & z < -z_{at} \\ \mathcal{E}_f(z), & -z_{at} < z < -z_{ar} \\ 0, & -z_{ar} < z < z_{ar} \\ -\mathcal{E}_f(z), & z_{ar} < z < z_{at} \\ 0, & z > z_{at} \end{cases} \quad (1.8.48)$$

where $\mathcal{E}_f$ is the local section flap effectiveness.

Using the definition of θ from Eq. (1.8.3) together with Eq. (1.8.44) in the relation for the Fourier coefficients, A_n , specified by Eq. (1.8.4), gives

$$\sum_{n=1}^N A_n \left[\frac{4b}{\widetilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = (\alpha - \alpha_{L0})_{\text{root}} - \Omega \omega(\theta) + \delta_a \chi(\theta) + \bar{p} \cos(\theta) \quad (1.8.49)$$

where $\bar{p}$ is a dimensionless rolling rate defined as $\bar{p} = pb/2V_\infty$. The Fourier coefficients in Eq. (1.8.49) can be conveniently written as

$$A_n = a_n(\alpha - \alpha_{L0})_{\text{root}} - b_n \Omega + c_n \delta_a + d_n \bar{p} \quad (1.8.50)$$

where

$$\sum_{n=1}^N a_n \left[\frac{4b}{\widetilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = 1 \quad (1.8.51)$$

$$\sum_{n=1}^N b_n \left[\frac{4b}{\widetilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = \omega(\theta) \quad (1.8.52)$$

$$\sum_{n=1}^N c_n \left[\frac{4b}{\widetilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = \chi(\theta) \quad (1.8.53)$$

$$\sum_{n=1}^N d_n \left[\frac{4b}{\widetilde{C}_{L,\alpha} c(\theta)} + \frac{n}{\sin(\theta)} \right] \sin(n\theta) = \cos(\theta) \quad (1.8.54)$$

Equations (1.8.51) and (1.8.52) are exactly Eqs. (1.8.20) and (1.8.21) and their solutions are obtained following Examples 1.8.1 and 1.8.2. The solutions to Eqs. (1.8.53) and (1.8.54) can be obtained in a similar manner and are both independent of angle of attack and rolling rate.

Once the Fourier coefficients are determined from Eqs. (1.8.50) through (1.8.54), the spanwise circulation distribution is known from Eq. (1.8.3) and the spanwise section lift distribution is given by

$$\tilde{L}(z) = \rho V_\infty \Gamma(z)$$

Thus, in view of Eq. (1.8.3), the rolling moment coefficient can be evaluated from

$$C_\ell = \frac{1}{\frac{1}{2} \rho V_\infty^2 S b} \int_{z=-b/2}^{b/2} \tilde{L}(z) z dz = \frac{2}{V_\infty S b} \int_{z=-b/2}^{b/2} \Gamma(z) z dz = -\frac{b^2}{S} \sum_{n=1}^N A_n \int_{\theta=0}^{\pi} \sin(n\theta) \cos(\theta) \sin(\theta) d\theta$$

or after applying the trigonometric identity, $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$, along with the definition of aspect ratio

$$C_\ell = -\frac{R_A}{2} \sum_{n=1}^N A_n \int_{\theta=0}^{\pi} \sin(n\theta) \sin(2\theta) d\theta \quad (1.8.55)$$

The integral in Eq. (1.8.55) is evaluated from

$$\int_{\theta=0}^{\pi} \sin(m\theta) \sin(n\theta) d\theta = \begin{cases} 0, & n \neq m \\ \pi/2, & n = m \end{cases} \quad (1.8.56)$$

After applying Eqs. (1.8.50) and (1.8.56), Eq. (1.8.55) becomes

$$C_\ell = -\frac{\pi R_A}{4} A_2 = -\frac{\pi R_A}{4} [a_2(\alpha - \alpha_{L0})_{\text{root}} - b_2 \Omega + c_2 \delta_a + d_2 \bar{p}] \quad (1.8.57)$$

For a wing with a **spanwise symmetric planform and spanwise symmetric washout**, the solutions to Eqs. (1.8.51) and (1.8.52) give

$$a_n = b_n = 0, \quad n \text{ even} \quad (1.8.58)$$

and Eq. (1.8.57) reduces to

$$C_\ell = C_{\ell, \delta_a} \delta_a + C_{\ell, \bar{p}} \bar{p} \quad (1.8.59)$$

where

$$C_{\ell, \delta_a} = -\frac{\pi R_A}{4} c_2 \quad (1.8.60)$$

$$C_{\ell, \bar{p}} = -\frac{\pi R_A}{4} d_2 \quad (1.8.61)$$

The Fourier coefficients obtained from Eq. (1.8.53) depend on control surface geometry as well as the planform shape of the wing. Thus, the change in rolling moment coefficient with respect to aileron deflection depends on the size and shape of the ailerons

and the wing planform. On the other hand, the Fourier coefficients that are evaluated from Eq. (1.8.54) are functions of only wing planform. For an elliptic planform, the spanwise variation in chord length is given by

$$c(y) = \frac{4b}{\pi R_A} \sqrt{1 - (2y/b)^2} \quad \text{or} \quad c(\theta) = \frac{4b}{\pi R_A} \sin(\theta)$$

and Eq. (1.8.54) reduces to

$$\sum_{n=1}^N d_n \left(\frac{\pi R_A}{\tilde{C}_{L,\alpha}} + n \right) \sin(n\theta) = \sin(\theta) \cos(\theta) \quad (1.8.62)$$

The solution to Eq. (1.8.62) is given by the Fourier integral

$$d_n \left(\frac{\pi R_A}{\tilde{C}_{L,\alpha}} + n \right) = \frac{2}{\pi} \int_0^\pi \sin(\theta) \cos(\theta) \sin(n\theta) d\theta = \frac{1}{\pi} \int_0^\pi \sin(2\theta) \sin(n\theta) d\theta \quad (1.8.63)$$

which is readily evaluated from Eq. (1.8.56) to give

$$d_n = \begin{cases} \frac{\tilde{C}_{L,\alpha}}{2(\pi R_A + 2\tilde{C}_{L,\alpha})}, & n = 2 \\ 0, & n \neq 2 \end{cases} \quad (1.8.64)$$

Using Eq. (1.8.64) in Eq. (1.8.61) gives

$$C_{\ell, \bar{p}} = \frac{-\tilde{C}_{L,\alpha}}{8(1 + 2\tilde{C}_{L,\alpha}/\pi R_A)}$$

Thus, in view of Eq. (1.8.7), **the change in rolling moment coefficient with respect to dimensionless rolling rate for an elliptic wing** can be written as

$$C_{\ell, \bar{p}} = -\frac{\kappa_{\ell \bar{p}} C_{L,\alpha}}{8}, \quad \kappa_{\ell \bar{p}} \equiv \frac{1 + \tilde{C}_{L,\alpha}/\pi R_A}{1 + 2\tilde{C}_{L,\alpha}/\pi R_A} \quad (1.8.65)$$

Similarly, in the general case of **a wing with arbitrary planform**,

$$C_{\ell, \bar{p}} = -\frac{\kappa_{\ell \bar{p}} C_{L,\alpha}}{8}, \quad \kappa_{\ell \bar{p}} \equiv \frac{2d_2}{a_1} \quad (1.8.66)$$

Using an airfoil lift slope of 2π to determine a_1 and d_2 from Eqs. (1.8.51) and (1.8.54), the results shown in Fig. 1.8.25 are obtained for tapered wings.

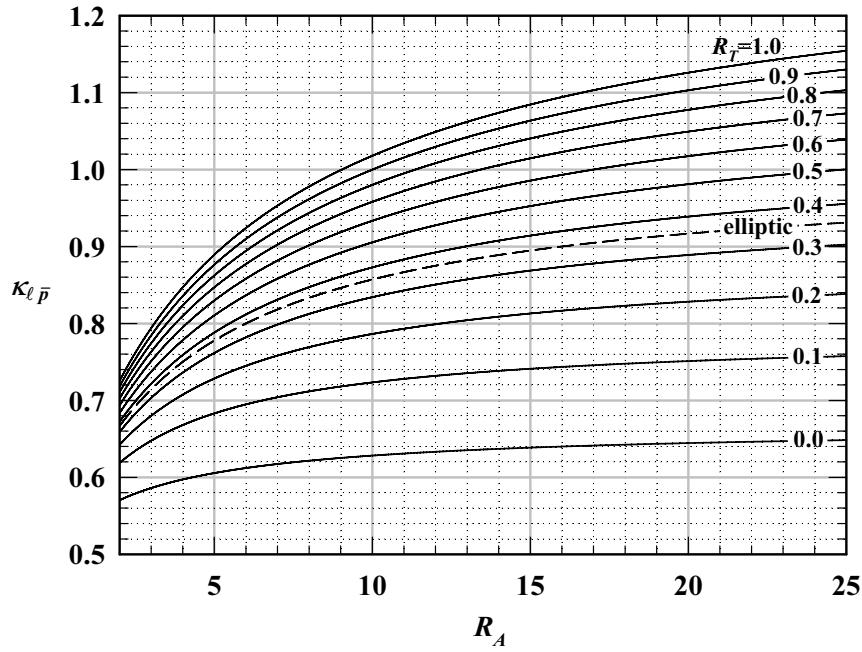


Figure 1.8.25. Roll damping factor for wings with linear taper from Eq. (1.8.66).

The asymmetric spanwise variations in aerodynamic angle of attack, which result from aileron deflection and roll, can also produce a yawing moment. The aerodynamic yawing moment is given the symbol n , and the traditional sign convention for yaw is positive to the right (i.e., a moment that would yaw the airplane's nose to the right). The yawing moment develops as a direct result of an asymmetric spanwise variation in drag. Thus, the yawing moment coefficient can be written as

$$C_n \equiv \frac{n}{\frac{1}{2} \rho V_\infty^2 S b} = \frac{1}{\frac{1}{2} \rho V_\infty^2 S b} \int_{z=-b/2}^{b/2} (-z) \tilde{D}(z) dz \quad (1.8.67)$$

As shown in Figs. 1.8.6 and 1.8.24, the asymmetric section drag results from tilting the section lift vector through the induced angle, α_i , and the roll angle, zp/V_∞ . Thus, after expressing the section lift in terms of the section circulation, we have

$$\tilde{D}(z) = \tilde{L} \sin(\alpha_i + zp/V_\infty) \cong \tilde{L}(\alpha_i + zp/V_\infty) = \rho V_\infty \Gamma(z) [\alpha_i(z) + zp/V_\infty] \quad (1.8.68)$$

The induced angle of attack as predicted by lifting-line theory is

$$\alpha_i(z) = \frac{1}{4\pi V_\infty} \int_{\zeta=-b/2}^{b/2} \frac{1}{z-\zeta} \left(\frac{d\Gamma}{d\zeta} \right)_{\zeta=\zeta} d\zeta \quad (1.8.69)$$

Using Eqs. (1.8.68) and (1.8.69) in Eq. (1.8.67) gives

$$C_n = \int_{z=-b/2}^{b/2} \frac{-2z\Gamma(z)}{V_\infty^2 S b} \left[\frac{1}{4\pi} \int_{\zeta=-b/2}^{b/2} \frac{1}{z-\zeta} \left(\frac{d\Gamma}{dz} \right)_{z=\zeta} d\zeta + zp \right] dz \quad (1.8.70)$$

From Eq. (1.8.3) we have

$$\Gamma(\theta) = 2bV_\infty \sum_{n=1}^N A_n \sin(n\theta), \quad \cos(\theta) = -\frac{2z}{b}, \quad \frac{d\Gamma}{dz} = 4V_\infty \sum_{n=1}^N A_n \frac{n \cos(n\theta)}{\sin(\theta)}$$

and Eq. (1.8.70) can be written as

$$C_n = R_A \int_{\theta=0}^{\pi} \cos(\theta) \left[\sum_{n=1}^N A_n \sin(n\theta) \right] \left[\sum_{n=1}^N \frac{nA_n}{\pi} \int_{\phi=0}^{\pi} \frac{\cos(n\phi)d\phi}{\cos(\phi) - \cos(\theta)} \right] \sin(\theta) d\theta \\ - R_A \bar{p} \sum_{n=1}^N A_n \int_{\theta=0}^{\pi} \cos^2(\theta) \sin(\theta) \sin(n\theta) d\theta$$

Integrating in ϕ and using the trigonometric relation, $\sin(2\theta) = 2\cos(\theta)\sin(\theta)$, yields

$$C_n = R_A \sum_{n=1}^N \sum_{m=1}^N nA_n A_m \int_{\theta=0}^{\pi} \cos(\theta) \sin(n\theta) \sin(m\theta) d\theta \\ - R_A \bar{p} \sum_{n=1}^N A_n \frac{1}{2} \int_{\theta=0}^{\pi} \cos(\theta) \sin(2\theta) \sin(n\theta) d\theta \quad (1.8.71)$$

Since m and n are positive integers, the integrals with respect to θ are evaluated from

$$\int_{\theta=0}^{\pi} \cos(\theta) \sin(n\theta) \sin(m\theta) d\theta = \begin{cases} \pi/4, & m = n \pm 1 \\ 0, & m \neq n \pm 1 \end{cases}$$

and after some rearranging, we obtain

$$C_n = \frac{\pi R_A}{4} \sum_{n=2}^N (2n-1) A_{n-1} A_n - \frac{\pi R_A \bar{p}}{8} (A_1 + A_3) \quad (1.8.72)$$

After applying Eq. (1.8.5), we find that the **yawing moment coefficient for a wing of arbitrary geometry** is given by

$$C_n = \frac{C_L}{8} (6A_2 - \bar{p}) + \frac{\pi R_A}{8} (10A_2 - \bar{p}) A_3 + \frac{\pi R_A}{4} \sum_{n=4}^N (2n-1) A_{n-1} A_n \quad (1.8.73)$$

where the Fourier coefficients, A_n , are related to Ω , δ_a , and $\bar{p}$ through Eq. (1.8.50).

If the wing planform and washout distribution are both spanwise symmetric, all even coefficients in both a_n and b_n are zero. The change in aerodynamic angle of attack that results from roll is always a spanwise odd function. Thus, all odd coefficients in d_n are zero for a spanwise symmetric wing. If the ailerons produce an equal and opposite change on each semispan, the control surface distribution function is a spanwise odd function as well and all odd coefficients in c_n are also zero. With this **wing symmetry** the Fourier coefficients must satisfy the relations $a_n = b_n = 0$ for n even and $c_n = d_n = 0$ for n odd. Thus, Eq. (1.8.50) reduces to

$$A_n = \begin{cases} a_n(\alpha - \alpha_{L0})_{\text{root}} - b_n\Omega, & n \text{ odd} \\ c_n\delta_a + d_n\bar{p}, & n \text{ even} \end{cases} \quad (1.8.74)$$

Using the Fourier coefficients from Eq. (1.8.74) in Eq. (1.8.73), the **yawing moment coefficient with the wing symmetry described above** is given by

$$\begin{aligned} C_n = & \frac{C_L}{8} [6c_2\delta_a - (1 - 6d_2)\bar{p}] + \frac{\pi R_A}{8} [10c_2\delta_a - (1 - 10d_2)\bar{p}] [a_3(\alpha - \alpha_{L0})_{\text{root}} - b_3\Omega] \\ & + \frac{\pi R_A}{4} \left[\sum_{n=4}^N (2n-1) [a_{n-1}(\alpha - \alpha_{L0})_{\text{root}} - b_{n-1}\Omega] (c_n\delta_a + d_n\bar{p}) \right]_{n \text{ even}} \\ & + \frac{\pi R_A}{4} \left[\sum_{n=5}^N (2n-1) (c_{n-1}\delta_a + d_{n-1}\bar{p}) [a_n(\alpha - \alpha_{L0})_{\text{root}} - b_n\Omega] \right]_{n \text{ odd}} \end{aligned} \quad (1.8.75)$$

For the special case of a **symmetric wing operating with optimum washout**, which is specified by Eq. (1.8.42), $a_n(\alpha - \alpha_{L0})_{\text{root}} - b_n\Omega$ is always zero for $n > 1$ and

$$C_n = C_{n,\delta_a}\delta_a + C_{n,\bar{p}}\bar{p}, \quad C_{n,\delta_a} = \frac{3C_L}{4}c_2, \quad C_{n,\bar{p}} = -\frac{C_L}{8}(1 - 6d_2) \quad (1.8.76)$$

For wings without optimum washout, the higher-order terms in Eq. (1.8.75) are not too large and the linear relation from Eq. (1.8.76) can be used as a rough approximation.

By comparing Eq. (1.8.59) with Eq. (1.8.76) we find that **within the accuracy of Eq. (1.8.76)**, the yawing moment can be expressed in terms of the rolling moment and the rolling rate,

$$C_n = -\frac{3C_L}{\pi R_A}C_\ell - \frac{C_L}{8}\bar{p} \quad (1.8.77)$$

When an airplane responds to aileron input, the rolling rate quickly reaches the steady-state value, which results in no rolling moment. Thus, we see from Eq. (1.8.77) that the magnitude of the steady yawing moment produced by aileron deflection is proportional to the lift coefficient developed by the wing and the steady rolling rate that develops as a

result of the aileron deflection. From Eq. (1.8.77), we can also see that the sign of the yawing moment, which is induced on a wing by aileron deflection, is opposite to that of the rolling moment induced by the same aileron deflection. Positive aileron deflection induces a wing rolling moment to the left and a wing yawing moment to the right. For this reason, the yawing moment induced on the wing by aileron deflection is commonly called *adverse yaw*. We can also see from Eq. (1.8.77) that adverse yaw is more pronounced at low airspeeds, which require higher values of C_L .

EXAMPLE 1.8.3. The rectangular wing of aspect ratio 8.0, which was described in Example 1.8.1, has ailerons with sealed hinges that extend from the spanwise coordinate $z/b = \pm 0.25$ to $z/b = \pm 0.45$. The aileron chord length is constant and equal to 18 percent of the wing chord. Estimate the rolling and yawing moment coefficients that would be induced on this wing by a 5-degree deflection of these ailerons when the wing is operating at the design lift coefficient of 0.4 and there is no rolling rate. What steady dimensionless rolling rate could be sustained by this aileron deflection?

Solution. Since $c_f/c = 0.18$, from Fig. 1.7.4, the hinge efficiency for these sealed flaps is estimated to be 0.85. Because the flap deflection is less than 10 degrees, from Fig. 1.7.5, a deflection efficiency of 1.0 is used. Thus, by using Eqs. (1.7.5) and (1.7.6) in Eq. (1.7.13), the section flap effectiveness for these ailerons is constant and given by

$$\theta_f = \cos^{-1} \left(2 \frac{c_f}{c} - 1 \right) = 2.265, \quad \varepsilon_f = (0.85)(1.0) \left[1 - \frac{\theta_f - \sin(\theta_f)}{\pi} \right] = 0.445$$

With this result, the control surface distribution function from Eq. (1.8.48) is

$$\chi(z) \equiv \begin{cases} 0.000, & z/b < -0.45 \\ 0.445, & -0.45 < z/b < -0.25 \\ 0.000, & -0.25 < z/b < 0.25 \\ -0.445, & 0.25 < z/b < 0.45 \\ 0.000, & z/b > 0.45 \end{cases}$$

In view of Eq. (1.8.53), the Fourier coefficients, c_n , are obtained from

$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & \cdots & C_{1N} \\ C_{21} & C_{22} & C_{23} & \cdots & C_{2N} \\ C_{31} & C_{32} & C_{33} & \cdots & C_{3N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ C_{N1} & C_{N2} & C_{N3} & \cdots & C_{NN} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_N \end{bmatrix} = \begin{bmatrix} \chi(\theta_1) \\ \chi(\theta_2) \\ \chi(\theta_3) \\ \vdots \\ \chi(\theta_N) \end{bmatrix}$$

where $[C]$ is the same matrix as that used in Example 1.8.1. For $N=99$, this gives

$$\begin{Bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \\ \vdots \\ c_{98} \\ c_{99} \end{Bmatrix} = [\mathbf{C}]^{-1} \begin{Bmatrix} \chi(\theta_1) \\ \chi(\theta_2) \\ \chi(\theta_3) \\ \chi(\theta_4) \\ \vdots \\ \chi(\theta_{98}) \\ \chi(\theta_{99}) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0.03853294 \\ 0 \\ 0.00335119 \\ \vdots \\ -0.00001777 \\ 0 \end{Bmatrix}$$

From Eq. (1.8.60), the change in the rolling moment coefficient with respect to aileron deflection is

$$C_{\ell, \delta_a} = -\frac{\pi R_A}{4} c_2 = -\frac{\pi(8.0)}{4} 0.0385 = -0.242$$

and from Eq. (1.8.59) with no rolling rate,

$$C_\ell = C_{\ell, \delta_a} \delta_a = -0.242 \frac{5.0\pi}{180} = \underline{-0.0211}$$

Since this wing has optimum washout for this lift coefficient, from Eq. (1.8.77),

$$C_n = -\frac{3C_L}{\pi R_A} C_\ell = -\frac{3(0.4)}{\pi(8.0)} (-0.0211) = \underline{0.00101}$$

This result applies only for the case of optimum washout, which in this case requires 4.64 degrees of elliptic washout. For the same wing planform with no washout, carrying the higher-order terms using either Eq. (1.8.73) or (1.8.75) gives $C_n = 0.00123$, and with 4.5 degrees of linear washout we get $C_n = 0.00087$.

Similarly, the coefficients, d_n , are evaluated from Eq. (1.8.54) and the change in the rolling moment with respect to rolling rate is found from Eq. (1.8.61):

$$\begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ \vdots \\ d_{98} \\ d_{99} \end{Bmatrix} = [\mathbf{C}]^{-1} \begin{Bmatrix} \cos(\theta_1) \\ \cos(\theta_2) \\ \cos(\theta_3) \\ \cos(\theta_4) \\ \vdots \\ \cos(\theta_{98}) \\ \cos(\theta_{99}) \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0.09411716 \\ 0 \\ 0.01326130 \\ \vdots \\ 0.00000241 \\ 0 \end{Bmatrix}, \quad C_{\ell, \bar{p}} = -\frac{\pi R_A}{4} d_2 = -0.591$$

Setting the net rolling moment coefficient to zero, Eq. (1.8.59) is solved for the steady dimensionless rolling rate:

$$\bar{p}_{\text{steady}} = -(C_{\ell, \delta_a} / C_{\ell, \bar{p}}) \delta_a = -(-0.242 / -0.591)(5.0\pi / 180) = \underline{-0.0357}$$

Wing Aspect Ratio and Geometric Mean Chord Length

We have seen that the aspect ratio of a finite wing has a profound effect on the wing's performance. As defined in Eq. (1.8.5), the aspect ratio for a wing of arbitrary planform can be computed as the square of the wingspan divided by the planform area,

$$R_A \equiv \frac{b^2}{S} \quad (1.8.78)$$

In a more general sense, aspect ratio is defined to be the ratio of the longer to the shorter dimension for any two-dimensional shape. The aspect ratio of a rectangle is simply the length of the long side divided by the length of the short side. Thus, the **aspect ratio of a rectangular wing** having a constant chord, c , and wingspan, b , can be defined as

$$R_A \equiv \frac{b}{c} \quad (1.8.79)$$

Because the planform area of a rectangular wing is simply the wingspan multiplied by the chord length, $S = bc$, the definitions given in Eqs. (1.8.78) and (1.8.79) are equivalent. For the more general case of a wing having a chord length that varies with the spanwise coordinate, z , the aspect ratio can be written as the ratio of the wingspan to the average or *mean chord length*, $\bar{c}$,

$$R_A \equiv \frac{b}{\bar{c}} \quad (1.8.80)$$

where, by equating the definitions of aspect ratio in Eqs. (1.8.78) and (1.8.80), we see that the **mean chord length** is defined as

$$\bar{c} \equiv \frac{S}{b} = \frac{1}{b} \int_{z=-b/2}^{b/2} c(z) dz \quad (1.8.81)$$

Because the wing aspect ratio defined by Eq. (1.8.78) has such a profound effect on wing performance, **both the wingspan and the mean chord length defined by Eq. (1.8.81) are important characteristic length scales for a finite wing.**

For a wing with linear taper, the planform area is one half the sum of the root chord and the tip chord multiplied by the wingspan. Thus, the mean chord length for a **wing with linear taper** is

$$\bar{c} = \frac{1}{b} \int_{z=-b/2}^{b/2} \left[c_{\text{root}} - (c_{\text{root}} - c_{\text{tip}}) \left| \frac{2z}{b} \right| \right] dz = \frac{c_{\text{root}} + c_{\text{tip}}}{2} = \frac{1 + R_T}{2} c_{\text{root}} \quad (1.8.82)$$

For an **elliptic wing** the mean chord length is

$$\bar{c} = \frac{1}{b} \int_{z=-b/2}^{b/2} c_{\text{root}} \sqrt{1 - \left(\frac{2z}{b} \right)^2} dz = \frac{\pi}{4} c_{\text{root}} \quad (1.8.83)$$

Wing Camber and Mean Aerodynamic Chord

Because section lift does not contribute to the pitching moment about the lifting line of an unswept wing, the pitching moment coefficient about the wing's lifting line depends only on wing camber. From the definitions of wing and airfoil moment coefficients, the pitching moment coefficient about the origin of the coordinate system in Fig. 1.8.5 is

$$C_{m_0} = \frac{1}{S c_{ref}} \int_{z=-b/2}^{b/2} \tilde{C}_{m_{ac}} c^2 dz \quad (1.8.84)$$

where $\tilde{C}_{m_{ac}}$ is the section moment coefficient about the section aerodynamic center and c_{ref} is the reference chord length used to define the wing pitching moment coefficient. For **wings with no control surface deflection or any other form of aerodynamic twist**, the section moment coefficient is constant over the wingspan and Eq. (1.8.84) becomes

$$C_{m_0} = \frac{\tilde{C}_{m_{ac}}}{S c_{ref}} \int_{z=-b/2}^{b/2} c^2 dz \quad (1.8.85)$$

Equation (1.8.85) leads to the definition of a characteristic length scale associated with wing camber, which is typically referred to as the **mean aerodynamic chord**

$$\bar{c}_{mac} \equiv \frac{1}{S} \int_{z=-b/2}^{b/2} c^2 dz \quad (1.8.86)$$

For **wings with linear taper** the mean aerodynamic chord and its location are given by

$$\bar{c}_{mac} = \frac{2}{3} \frac{1 + R_T + R_T^2}{1 + R_T} c_{root}, \text{ located at } \frac{z_{mac}}{b} = \frac{1}{6} \frac{1 + 2R_T}{1 + R_T} \quad (1.8.87)$$

For **wings of elliptic planform** Eq. (1.8.86) results in

$$\bar{c}_{mac} = \frac{8}{3\pi} c_{root}, \text{ located at } \frac{z_{mac}}{b} = \frac{1}{6\pi} \sqrt{9\pi^2 - 64} \quad (1.8.88)$$

Both $\bar{c}$ and $\bar{c}_{mac}$ are commonly used as the reference length for defining the wing pitching moment coefficient. In this textbook $\bar{c}$ is used for this purpose. This choice is completely arbitrary. To convert any moment coefficient based on $\bar{c}$ to that based on $\bar{c}_{mac}$, we simply multiply by the ratio $\bar{c}/\bar{c}_{mac}$, which is typically slightly less than unity. For example, the geometric relations for **wings with linear taper** produce the result

$$\bar{c}/\bar{c}_{mac} = 3(1 + R_T)^2 / [4(1 + R_T + R_T^2)] \quad (1.8.89)$$

which for taper ratios of 1.0 and 0.5 yields $\bar{c}/\bar{c}_{mac} = 1.00$ and $\bar{c}/\bar{c}_{mac} \cong 0.96$, respectively. For the **elliptic planform** the ratio $\bar{c}/\bar{c}_{mac}$ is

$$\bar{c}/\bar{c}_{mac} = 3\pi^2/32 \cong 0.93 \quad (1.8.90)$$

The Basic and Additional Section Lift Distributions

For many applications in aircraft design it is important to know how the lift is distributed over the span of a wing. From Eq. (1.8.3) combined with the vortex lifting law, lifting-line theory predicts that the spanwise variation in section lift for an unswept wing is

$$\tilde{L}(\theta) = \rho V_\infty \Gamma(\theta) = 2b\rho V_\infty^2 \sum_{n=1}^{\infty} A_n \sin(n\theta), \quad \theta = \cos^{-1}(-2z/b) \quad (1.8.91)$$

Thus, the spanwise variation in local section lift coefficient is given by

$$\tilde{C}_L(\theta) \equiv \frac{\tilde{L}(\theta)}{\frac{1}{2}\rho V_\infty^2 c(\theta)} = \frac{4b}{c(\theta)} \sum_{n=1}^{\infty} A_n \sin(n\theta) \quad (1.8.92)$$

The Fourier coefficients, A_n , in Eq. (1.8.92) can be written using the change of variables given by Eq. (1.8.19),

$$A_n \equiv a_n(\alpha - \alpha_{L0})_{\text{root}} - b_n \Omega$$

Furthermore, after applying Eqs. (1.8.26) and (1.8.28) to Eq. (1.8.24), we have

$$C_L = \pi R_A [a_1(\alpha - \alpha_{L0})_{\text{root}} - b_1 \Omega]$$

or after solving for the root aerodynamic angle of attack

$$(\alpha - \alpha_{L0})_{\text{root}} = \frac{b_1}{a_1} \Omega + \frac{C_L}{\pi R_A a_1} \quad (1.8.93)$$

Using Eq. (1.8.93) in Eq. (1.8.19) yields

$$A_n = \left(\frac{b_1 a_n}{a_1} - b_n \right) \Omega + \frac{a_n}{\pi R_A a_1} C_L \quad (1.8.94)$$

Using this change of variables, Eq. (1.8.92) can be written as

$$\tilde{C}_L(\theta) = \Omega \sum_{n=1}^{\infty} 4 \left(\frac{b_1 a_n}{a_1} - b_n \right) \frac{\sin(n\theta)}{c(\theta)/b} + C_L \sum_{n=1}^{\infty} \frac{4a_n}{\pi R_A a_1} \frac{\sin(n\theta)}{c(\theta)/b} \quad (1.8.95)$$

We see from Eq. (1.8.95) that the spanwise variation in local section lift coefficient can be divided conveniently into two components. The first term on the right-hand side of Eq. (1.8.95) is called the *basic section lift coefficient* and the second term is called the *additional section lift coefficient*. The basic section lift coefficient is independent of C_L and directly proportional to the total amount of wing twist, Ω . The additional section lift coefficient at any section of the wing is independent of wing twist and directly proportional to the net wing lift coefficient, C_L .

As can be seen from Eq. (1.8.95), **the basic section lift coefficient is the spanwise variation in local section lift coefficient that occurs when the total net lift developed by the wing is zero.** Examination of the first term on the right-hand side of Eq. (1.8.95) reveals that the basic section lift coefficient depends on all of the Fourier coefficients a_n and b_n . From Eq. (1.8.20) we see the Fourier coefficients a_n depend only on the wing planform. Equation (1.8.21) shows that the Fourier coefficients b_n depend on both the wing planform and the dimensionless twist distribution function, $\omega(\theta)$. Thus, **the spanwise variation in the basic section lift coefficient depends on wing planform and wing twist but is independent of the wing's angle of attack.**

Examination of the second term on the right-hand side of Eq. (1.8.95) discloses that the additional section lift coefficient depends only on the wing planform and the Fourier coefficients a_n . From Eq. (1.8.20) we have seen that the coefficients a_n do not depend on wing twist. Thus, for an unswept wing, Eq. (1.8.95) exposes the important fact that **the additional section lift coefficient is independent of wing twist.** Because the basic section lift coefficient is zero for an untwisted wing, we see that **the additional section lift coefficient is equivalent to the spanwise variation in local section lift coefficient that would be developed on an untwisted wing of the same planform operating at the same wing lift coefficient.**

Figure 1.8.26 shows how the net section lift coefficient and its two components obtained from Eq. (1.8.95) vary along the span of an unswept wing with linear taper of aspect ratio 8.0 and taper ratio 0.5. This figure shows the spanwise variation in wing section lift coefficient for several values of total linear twist with the net wing lift coefficient held constant at 1.0. Similar results are shown in Fig. 1.8.27 for three different values of wing lift coefficient with total linear twist held constant at 6 degrees. Notice that, whereas the center of total lift on each semispan of this wing moves inboard

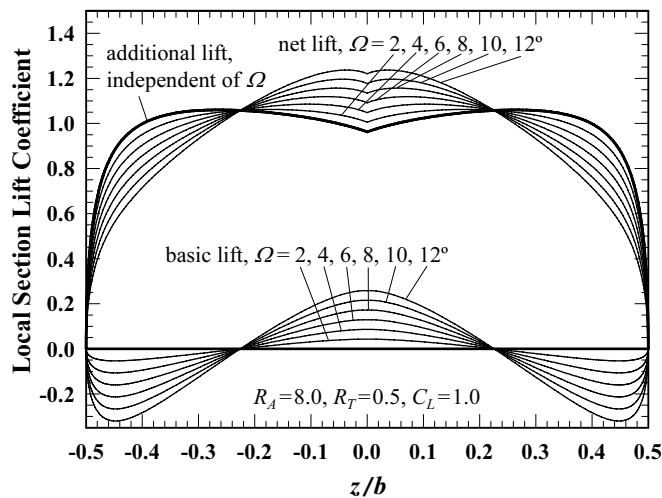


Figure 1.8.26. Spanwise variation in local section lift coefficient as a function of the total amount of linear twist with the net wing lift coefficient held constant at $C_L = 1.0$.

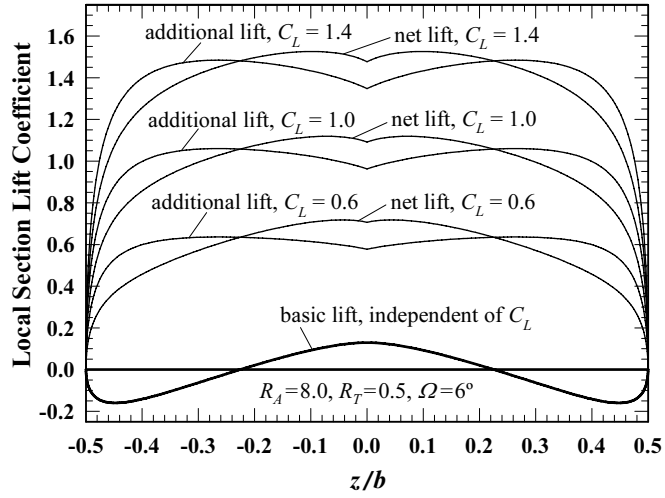


Figure 1.8.27. Spanwise variation in local section lift coefficient as a function of the net wing lift coefficient with the total amount of linear twist held constant at $\Delta = 6^\circ$.

as washout is increased, the center of additional lift on each semispan does not change with either the amount of wing twist or the net wing lift coefficient.

Here $\tilde{C}_{L_b}$ will be used to signify the **basic section lift coefficient**,

$$\tilde{C}_{L_b}(\theta) \equiv \Delta \sum_{n=1}^{\infty} 4 \left(\frac{b_1 a_n}{a_1} - b_n \right) \frac{\sin(n\theta)}{c(\theta)/b} \quad (1.8.96)$$

and $\tilde{C}_{L_a}$ will be used to denote the **additional section lift coefficient**,

$$\tilde{C}_{L_a}(\theta) \equiv C_L \sum_{n=1}^{\infty} \frac{4a_n}{\pi R_A a_1} \frac{\sin(n\theta)}{c(\theta)/b} \quad (1.8.97)$$

The total section lift coefficient at any spanwise section of the wing is simply the sum of the basic and additional section lift coefficients,

$$\tilde{C}_L(\theta) = \tilde{C}_{L_b}(\theta) + \tilde{C}_{L_a}(\theta) \quad (1.8.98)$$

The important points to remember regarding these two components of lift are:

1. *The basic section lift coefficient is independent of angle of attack.*
2. *The additional section lift coefficient is independent of wing twist.*
3. *Both the basic and additional section lift coefficients depend on wing planform.*

Semispan Aerodynamic Center and Moment Components

The spanwise distribution of section aerodynamic loads acting on each semispan of a finite wing can be replaced with a resultant force acting at the aerodynamic center of the semispan and a resultant moment that does not change with angle of attack. Because drag is typically small compared with the lift, drag is commonly neglected in estimating the position of the aerodynamic center. This traditional approximation is presented here. The effect of drag on the position of the aerodynamic center is considered in Chapter 4.

As a first approximation, the aerodynamic center of each wing semispan is sometimes assumed to be located at the section aerodynamic center of the airfoil section located at the spanwise coordinate of the semispan area centroid. Here the chord line that passes through the semispan area centroid is referred to as the *centroidal chord*. The spanwise coordinate of the wing semispan area centroid is given by

$$\bar{z}_c \equiv \frac{2}{S} \int_{z=0}^{b/2} c z dz \quad (1.8.99)$$

For **wings with constant linear taper**, i.e., trapezoidal wings, Eq. (1.8.99) results in

$$\frac{\bar{z}_c}{b} = \frac{1}{6} \frac{1 + 2R_T}{1 + R_T} \quad (1.8.100)$$

For **wings of elliptic planform**, the spanwise coordinate of the semispan centroid is

$$\frac{\bar{z}_c}{b} = \frac{2}{3\pi} \quad (1.8.101)$$

The location specified by Eq. (1.8.100) is commonly referred to as the location of the mean aerodynamic chord. Referring to the centroidal chord of a trapezoidal wing as the mean aerodynamic chord can be misleading, because it could be taken to imply that the location of the mean aerodynamic chord is significant for other wing geometries as well. However, the mean aerodynamic chord passes through the semispan centroid only for the special case of a trapezoidal wing. For example, the mean aerodynamic chord of an elliptic wing is located according to Eq. (1.8.88) at $z_{mac}/b \cong 0.264$, whereas the location of the centroidal chord is given by Eq. (1.8.101) as $\bar{z}_c/b \cong 0.212$.

In general, the semispan aerodynamic center of a wing is not located along either the centroidal chord or the mean aerodynamic chord. For example, Fig. 1.8.28 shows the aerodynamic center, centroidal chord, and mean aerodynamic chord for several different semispan geometries. Equation (1.8.99) gives the true spanwise location of the semispan aerodynamic center only if the additional section lift coefficient is uniform across the wingspan. Because a uniform additional section lift coefficient is produced by an elliptic wing with no sweep or dihedral in the locus of airfoil section aerodynamic centers, the semispan aerodynamic center of such wings is located along the centroidal chord as specified by Eq. (1.8.101). However, wings with linear taper do not produce a uniform additional section lift coefficient. **Thus, Eq. (1.8.100) should be used only as a rough estimate for the semispan aerodynamic center of a trapezoidal wing.**

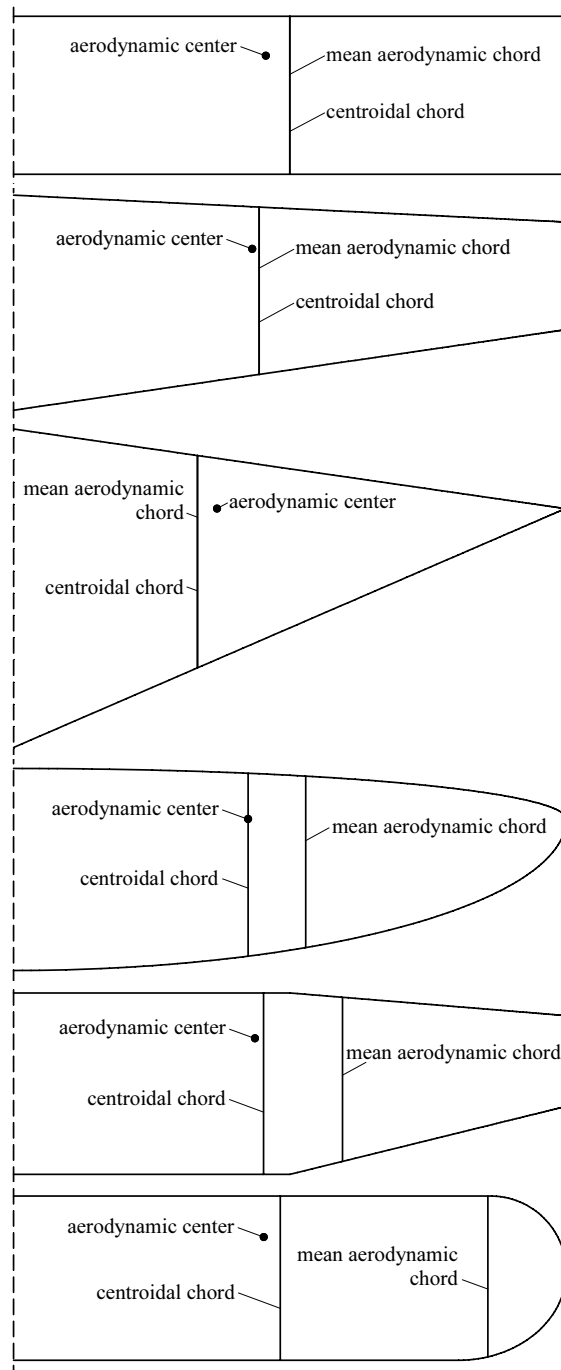


Figure 1.8.28. The aerodynamic center, centroidal chord, and mean aerodynamic chord for six different semispan geometries, all having the same aspect ratio and no quarter-chord sweep.

A more accurate estimate for the location of the semispan aerodynamic center of a wing with no sweep or dihedral in the locus of wing section aerodynamic centers can be obtained from Eqs. (1.8.96)–(1.8.98). Because we are neglecting drag, the resultant aerodynamic moment produced on each semispan of a wing about the origin of the coordinate system shown in Fig. 1.8.5 can be resolved into a pitching component about the z -axis and a rolling component about the freestream velocity vector.

Because section lift does not contribute to the pitching moment about the wing's lifting line, the contribution of the left semispan of an unswept wing to the pitching moment coefficient about the origin of the coordinate system shown in Fig. 1.8.5 is

$$(C_{m_0})_{\text{left}} = \frac{1}{S c_{\text{ref}}} \int_{z=0}^{b/2} \tilde{C}_{m_{ac}} c^2 dz \quad (1.8.102)$$

The moment coefficient specified by Eq. (1.8.102) is also the *root twisting moment coefficient* resulting from the aerodynamic load on this unswept wing. This moment results only from the effects of wing camber and is independent of geometric twist. For a wing with constant section pitching moment coefficient, Eq. (1.8.102) yields

$$(C_{m_0})_{\text{left}} = \frac{\bar{c}_{mac}}{c_{\text{ref}}} \frac{\bar{C}_{mac}}{2} \quad (1.8.103)$$

In a more general sense, Eq. (1.8.102) could be thought of in terms of a mean section pitching moment coefficient,

$$(C_{m_0})_{\text{left}} = \frac{\bar{c}_{mac}}{c_{\text{ref}}} \frac{\bar{\bar{C}}_{mac}}{2}, \quad \bar{\bar{C}}_{mac} \equiv \frac{2}{S \bar{c}_{mac}} \int_{z=0}^{b/2} \tilde{C}_{m_{ac}} c^2 dz \quad (1.8.104)$$

The contribution of the left wing semispan to the rolling moment coefficient about the origin of the coordinate system shown in Fig. 1.8.5 is

$$(C_{\ell_0})_{\text{left}} = \frac{1}{Sb} \int_{z=0}^{b/2} \tilde{C}_L c z dz = \frac{1}{Sb} \int_{z=0}^{b/2} (\tilde{C}_{L_b} + \tilde{C}_{L_a}) c z dz \quad (1.8.105)$$

It is important to note that, within the small angle approximation, the moment coefficient specified by Eq. (1.8.105) is the *root bending moment coefficient* resulting from the aerodynamic load on the wing semispan. Equating the distributed wing-section loading to a resultant force and moment acting at the aerodynamic center of the wing semispan, we can also write

$$(C_{\ell_0})_{\text{left}} = (C_{\ell_{ac}})_{\text{left}} + \frac{\bar{z}_{ac}}{Sb} \int_{z=0}^{b/2} \tilde{C}_L c dz = (C_{\ell_{ac}})_{\text{left}} + \frac{\bar{z}_{ac}}{b} \frac{C_L}{2} \quad (1.8.106)$$

where $(C_{\ell_{ac}})_{\text{left}}$ is the left wing semispan rolling moment coefficient about the semispan aerodynamic center and $\bar{z}_{ac}$ is the z -coordinate of the semispan aerodynamic center. Combining Eqs. (1.8.105) and (1.8.106) to eliminate the moment about the origin yields

$$(C_{\ell_{ac}})_{\text{left}} + \frac{\bar{z}_{ac}}{b} \frac{C_L}{2} = \frac{1}{Sb} \int_{z=0}^{b/2} (\tilde{C}_{L_b} + \tilde{C}_{L_a}) c z dz \quad (1.8.107)$$

Because the resultant moment about the aerodynamic center is invariant to small changes in angle of attack, differentiating Eq. (1.8.107) with respect to angle of attack, applying Eqs. (1.8.96) and (1.8.97), and solving for $\bar{z}_{ac}/b$ gives

$$\begin{aligned} \frac{\bar{z}_{ac}}{b} &= \frac{2}{SbC_{L,\alpha}} \frac{\partial}{\partial \alpha} \int_{z=0}^{b/2} (\tilde{C}_{L_b} + \tilde{C}_{L_a}) c z dz = \frac{2}{SbC_{L,\alpha}} \frac{\partial}{\partial \alpha} \int_{z=0}^{b/2} \tilde{C}_{L_a} c z dz \\ &= \frac{-2}{\pi} \sum_{n=1}^{\infty} \frac{a_n}{a_1} \int_{\theta=\pi/2}^{\pi} \sin(n\theta) \cos \theta \sin \theta d\theta = \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{a_n}{a_1} \int_{\theta=\pi}^{\pi/2} \sin(n\theta) \sin(2\theta) d\theta \end{aligned} \quad (1.8.108)$$

Because the additional section lift coefficient is independent of wing twist, Eq. (1.8.108) discloses the important fact that **the spanwise position of the aerodynamic center of the wing semispan is not affected by wing twist**. Recognizing that the even terms in a_n are always zero for spanwise symmetric wings, the integration in Eq. (1.8.108) yields

$$\frac{\bar{z}_{ac}}{b} = \frac{2}{3\pi} + \frac{2}{\pi} \sum_{n=3}^{\infty} \frac{\sin[(n-2)\pi/2]}{n^2-4} \frac{a_n}{a_1} = \frac{2}{3\pi} \left(1 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1} 3}{4n^2+4n-3} \frac{a_{2n+1}}{a_1} \right) \quad (1.8.109)$$

Figure 1.8.29 shows how $\bar{z}_{ac}/b$ varies with taper ratio and aspect ratio for wings with linear taper. The reader should note from Fig. 1.8.29 that, except for the special case of trapezoidal wings with a taper ratio near 0.35, **the spanwise location of the mean aerodynamic chord is not the semispan aerodynamic center**, as is commonly stated or inferred in the literature. For example, with a linear taper ratio of 0.8 the mean aerodynamic chord is located at $z_{mac}/b \cong 0.24$, whereas the semispan aerodynamic center varies from about $\bar{z}_{ac}/b \cong 0.22$ to $\bar{z}_{ac}/b \cong 0.23$, depending on aspect ratio. Furthermore, the location of the semispan aerodynamic center also varies with wing sweep, whereas the spanwise location of the mean aerodynamic chord does not.

Using Eq. (1.8.97) in Eq. (1.8.108), it can be shown that the spanwise coordinate of the semispan aerodynamic center can also be expressed as

$$\bar{z}_{ac} = \frac{2}{SC_L} \int_{z=0}^{b/2} \tilde{C}_{L_a} c z dz \quad (1.8.110)$$

Using Eq. (1.8.110) in Eq. (1.8.105) yields

$$\begin{aligned} (C_{\ell_0})_{\text{left}} &= \frac{1}{Sb} \int_{z=0}^{b/2} \tilde{C}_{L_a} c z dz + \frac{1}{Sb} \int_{z=0}^{b/2} \tilde{C}_{L_b} c z dz \\ &= \frac{\bar{z}_{ac}}{b} \frac{C_L}{2} + \frac{1}{Sb} \int_{z=0}^{b/2} \tilde{C}_{L_b} c z dz \end{aligned} \quad (1.8.111)$$

It is worth noting that the infinite series defined in Eq. (1.8.113) is dominated by the first term. In the case of a rectangular wing of aspect ratio 6.0, carrying only the first term on the right-hand side of Eq. (1.8.113) yields $\kappa_{M\Omega} = 0.024338$, whereas carrying 400 or more terms in this infinite series produces $\kappa_{M\Omega} = 0.024986$. For typical washout distributions $\kappa_{M\Omega}$ is positive. Figure 1.8.30 shows how $\kappa_{M\Omega}$ varies with taper ratio and aspect ratio for wings with constant linear taper and constant linear twist.

As expressed in Eq. (1.8.112), the root wing bending moment coefficient is composed of two components. The first is proportional to the semispan lift acting through a moment arm of $\bar{z}_{ac}$ and the second is proportional to the product of the wing lift slope and the wing twist. For a given wing planform, the value of the proportionality constant $\kappa_{M\Omega}$ depends on the way in which the twist is distributed along the wingspan. This dependence enters into Eq. (1.8.113) through the Fourier coefficients b_n , which depend on the twist distribution through Eq. (1.8.21). For the typical case where washout is greatest near the wingtips, $\kappa_{M\Omega}$ is positive as shown for the case of linear washout in Fig. 1.8.30. Thus, as might be expected, Eq. (1.8.112) shows that the root bending moment decreases linearly as washout is added at the wingtips. If a twist distribution were used that had the greatest washout near the wing root, then $\kappa_{M\Omega}$ would be negative and the root bending moment would increase in proportion to the amount of twist.

For a given planform and twist distribution, Eq. (1.8.112) shows that the change in bending moment with respect to Ω is directly proportional to the wing lift slope. This should be expected because $C_{L,\alpha}$ is a measure of the wing's lift response to a change in any aerodynamic angle, i.e., α , α_{L0} , or Ω . The lift slope for an unswept wing of arbitrary planform is given by Eq. (1.8.26) and is not affected by wing twist.

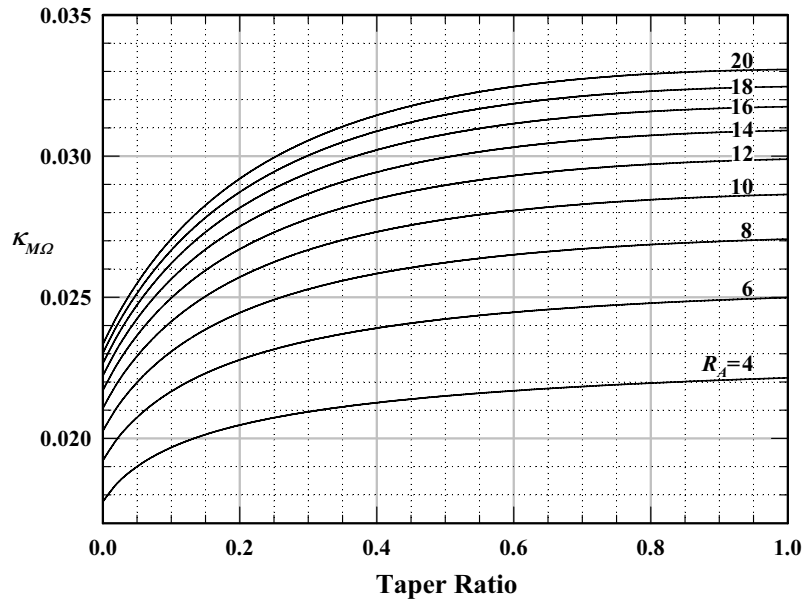


Figure 1.8.30. Twist factor in relations for the root wing bending moment coefficient.

An Alternative View of Downwash

From Prandtl's lifting-line theory we have seen that a finite lifting wing results in the generation of downwash. This downwash has a significant effect on the performance of the wing and other surfaces of an airplane. Mathematically we have described the downwash and its effects in terms of vorticity and circulation. However, the student may find it easier to understand the physics of downwash when it is described in terms of pressure and momentum.

When an airplane flies overhead, the downwash produced by the lifting wing could be measured on the ground if instruments of sufficient sensitivity were available. This downwash can be viewed as a result of the momentum imparted to the air by the force of the airplane's weight. The weight of the airplane is carried directly by the pressure difference between the upper and lower surfaces of the wing. However, the air also feels this pressure difference. The net effect of the pressure difference on the air is that a force is applied to the air equal to the airplane's weight. From Newton's second law, we know that such a force must result in a downward momentum imparted to the air. As the air directly beneath the airplane moves down, viscous forces from the adjacent air may slow its downward motion. However, these same viscous forces act on the adjacent air to impart downward momentum to that air as well. Newton's second law requires that the total downward momentum of the air cannot change unless acted upon by an external force. Ultimately, the Earth will stop this downward momentum, which began as the downwash from the airplane wing. The final manifestation of the downwash produced by a lifting wing is a very small pressure rise over a very large area of the Earth. The net effect of this pressure rise integrated over the total affected area is a force applied to the surface of the Earth equal to the airplane weight. Whether the airplane is resting on the ground or flying at 40,000 feet, the surface of the Earth must ultimately support the total weight of the aircraft. Because the airplane is moving rapidly over the ground, the downwash is spread over a large area of the Earth and usually goes undetected.

The vortex model that we have used to describe the effects of downwash on a finite wing is based on inviscid flow. This model predicts that the vorticity shed from the wing and the associated downwash will extend an infinite distance behind the wing. In reality, viscous effects in the air will eventually dissipate the wingtip vortices. However, the net downward momentum associated with these vortices does not change as a result of this viscous dissipation. No matter how large the viscosity, this downward momentum will eventually reach the ground, where it produces an aerodynamic force equal to the airplane's weight.

1.9. Flow over Multiple Lifting Surfaces

Prandtl's classical lifting-line equation, expressed in Eq. (1.8.2), applies only to a single lifting surface with no sweep or dihedral. In the development of this relation it was assumed that the only downwash induced on the wing was that induced by the trailing vortex sheet, which is shed from the wing itself. Airplanes are usually designed with more than one lifting surface. A typical configuration could be a wing combined with horizontal and vertical stabilizers. Each of these surfaces can generate lift and vorticity. The vorticity that is generated by one lifting surface will alter the flow about another.